

Disrupting Education? Experimental Evidence on Technology-Aided Instruction in India*

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Abstract

We present experimental evidence on the impact of a technology-aided after-school instruction program on learning outcomes in middle school grades in urban India, using a lottery that provided students with a voucher to cover program costs. A key feature of the program was its ability to individually customize educational content to match the level and rate of progress of each student. We find that lottery winners had large increases in test scores of 0.36σ in math and 0.22σ in Hindi over just a 4.5-month period. IV estimates suggest that attending the program for 90 days would increase math and Hindi test scores by 0.59σ and 0.36σ respectively. We find similar absolute test score gains for all students, but the relative gain was much greater for academically-weaker students because their rate of learning in the control group was close to zero. We show that the program was able to effectively cater to the very wide variation in student learning levels within a single grade by precisely targeting instruction to the level of student preparation. The program was highly cost-effective, both in terms of productivity per dollar and unit of time. Our results suggest that well-designed technology-aided instruction programs can sharply improve productivity in delivering education.

JEL codes: C93, I21, J24, O15

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1 Introduction

Developing countries have made impressive progress in improving school enrollment and completion in the last two decades. Yet, their productivity in converting education investments of time and money into human capital remains very low. For instance, in India, over 60% of children aged 6-14 cannot read at the second grade level, despite primary school enrollment rates over 95% (ASER, 2014). Further, there have been very limited improvements in learning outcomes in the past decade despite substantial increases in education spending in this period (Muralidharan, 2013). More generally, even in developed countries, productivity growth in the education sector lags the rest of the economy, perhaps because the ‘technology’ of schooling (classroom-based instruction) has changed very little over time compared to rapid technological progress in other fields (Bosworth, 2005; Pritchett, 2013).

Thus, it is not surprising that increasing the use of technology in instruction is seen as a leading candidate for ‘disrupting’ the status quo and improving productivity in education (Negroponte et al., 2006; Khan, 2012; Mead, 2016).¹ Yet, the evidence to date appears rather mixed: A recent review of evidence from high-quality studies on the impact of using technology in education globally reports “mixed evidence with a pattern of null results” (Bulman and Fairlie, 2016). Overall, the evidence thus far suggests that realizing the potential of technology-aided education to improve education will require paying careful attention to the *details* of the specific intervention, and the extent to which it alleviates binding constraints to learning.

In this paper, we present experimental evidence on the impact of a technology-led instructional program (called Mindspark) that aimed to leverage technology to improve education by paying sustained attention to such design details. Developed by a leading Indian education firm, the Mindspark software reflects over 10 years of product development; it has been used by over 400,000 students, has a database of over 45,000 test questions, and administers over a million questions across its users every day. A key feature of the software is its ability to use these data to finely benchmark the learning level of every student and dynamically customize the material being delivered to match the level and rate of progress made by each individual student. A second noteworthy feature is its ability to analyze these data to identify patterns of student errors, and precisely target content to alleviate conceptual ‘bottlenecks’ that may be difficult for teachers to diagnose or address at the individual student level in a classroom setting. Mindspark can be delivered in a variety of settings (in schools, in after-school centers, or through self-guided study); it is platform-agnostic (can be deployed through computers, tablets, or smartphones); and it can be used both online and offline.

¹A non-exhaustive list of posited channels of impact include using technology to consistently deliver high-quality content that may circumvent limitations in teachers’ own knowledge; delivering engaging (often game-based) interactive content that may improve student attention; delivering individually customized content for students; reducing the lag between students attempting a problem and receiving feedback; and, analyzing patterns of student errors to precisely target content to clarify specific areas of misunderstanding.

We evaluate the after-school Mindspark centers in this paper. The centers scheduled six days of instruction per week, for 90 minutes per day; each session was divided into 45 minutes of individual self-driven learning on the Mindspark software and 45 minutes of instructional support from a teaching assistant in groups of 12-15 students.² The centers aimed to serve students from low-income neighborhoods in Delhi, and charged a modest fee.³ Our evaluation was carried out in a sample of 619 students recruited for the study from public middle schools in Delhi. Around half of these students were randomly selected to receive a voucher offering free attendance at the centers. We measure program impacts using independently-conducted paper-and-pencil tests of student learning in math and Hindi (language) before and after the 4.5-month long intervention. These tests were linked using item response theory (IRT) to be comparable on a common scale across both rounds of testing and across different grades.

We report five main sets of results. First, the dynamic computer-based assessments allow us to provide a more granular description of student learning levels, their distribution within grades, and their evolution over grades than the literature to date. Using these data, we show that student learning levels in our sample are several grade-levels behind their grade-appropriate standard, and that this gap grows by grade. The average grade 6 student is around 2.5 grade levels below sixth grade standards in Math; by grade 9, this deficit increases to 4.5 grade levels. Thus, the default of classroom instruction based on grade-appropriate textbooks is likely to be considerably above the preparation level of most students (especially the lower-achieving ones). Consistent with this, we find that the absolute value-added on our independently-administered tests is close to zero for the bottom-third of students in the control group, and we cannot reject that these students made no academic progress through the school year.

Second, we find that students winning a program voucher scored 0.36σ higher in math and 0.22σ higher in Hindi relative to students who applied for but did not win the lottery. Relative to the control group, lottery winners experienced twice the test score value-added in math and 2.5 times that in Hindi during the study period of 4.5 months. These are intent-to-treat estimates reflecting an average attendance rate of 58%. Using the lottery as an instrumental variable for attendance (and additional assumptions discussed in Section 4.6), we estimate that attending the Mindspark centers for 90 days (which corresponds to 80% attendance for half a school year), would raise math and Hindi test scores by 0.59σ and 0.36σ respectively.

²The teaching assistant focused on helping students with completing homework and with exam preparation, while the instruction was mostly provided by the Mindspark software. Our results may therefore under-estimate the full-potential impact of 90-minutes of “blended” technology-aided learning because teacher time was not optimized for instruction (see sections 2.1.1 and 5.2 for details).

³The online and school-based models require fees that are not affordable for low-income families. The Mindspark centers were set up with philanthropic funding to make the platform more widely accessible, and were located in low-income neighborhoods. However, the funders preferred that a (subsidized) fee be charged, reflecting a widely-held view among donors that cost-sharing is necessary to avoid wasting subsidies on those who will not value or use the product (Cohen and Dupas, 2010). The subsidized fee of Rs. 200 per month (USD 3 per month) was benchmarked to that charged by providers of private tutoring in the vicinity.

Third, we find that the impact of winning a voucher does not vary significantly by baseline test scores, gender, or household socioeconomic status. Thus, consistent with the promise of customized technology-led instruction, the intervention was equally effective at improving test scores for *all* students. However, while the absolute impact was similar at all parts of the initial test score distribution, the *relative* impact was much greater for weaker students because the ‘business as usual’ rate of progress in the control group was close to zero for students in the lower third of the initial test score distribution.

Fourth, using detailed electronic records of every question presented to students in the treatment group by the Mindspark software, we document that: (a) there is considerable variation in students’ initial preparation for grade-appropriate work, with students enrolled in the same grade typically spanning five to six grade levels in their readiness; and (b) the software targets instruction very precisely to each student’s learning level, and updates this targeting in response to changes in student learning. Thus, the ability of the software to handle the heterogeneity in student preparedness spanning several grades appears to be an important (though not exclusive) mechanism of impact.

Fifth, the program was highly cost effective. The test score value-added in the treatment group (even based on ITT estimates) was over 100% greater than that in the control group, and was achieved at a lower expenditure per student than incurred in the public schooling system. Our results are particularly striking when considered in terms of productivity per unit of time. For instance, Muralidharan (2012) finds that providing individual-level performance bonuses to teachers in India led to test score gains of 0.54σ and 0.35σ in math and language for students exposed to the program for five years. This is one of the largest effect sizes seen to date in an experimental study on education in developing countries. Yet, we estimate that regularly attending Mindspark could yield similar gains in one tenth the time (half a year).

Our first contribution is to the literature on computer-aided learning (CAL), where the evidence to date has often been characterized as mixed (Bulman and Fairlie, 2016). Nevertheless, our reading of the evidence suggests that some clear patterns are starting to emerge (see Appendix B for our synthesis). Hardware-focused interventions that provide computers at home or at school seem to have no positive impact on learning outcomes (Angrist and Lavy, 2002; Barrera-Orsorio and Linden, 2009; Beuermann et al., 2015; Cristia et al., 2012; Malamud and Pop-Eleches, 2011).⁴ Pedagogy-focused CAL programs that allow students to review grade-appropriate content at their own pace do better, but the gains are modest and range from 0.1σ to 0.2σ .⁵ Finally, interventions that use technology to also personalize instruction seem to deliver substantial gains. For instance, Banerjee et al. (2007) find math

⁴These disappointing results are likely explained by the fact that hardware-focused interventions have done little to change instruction, and at times have crowded out student time for independent study.

⁵See, for example, Carrillo et al. (2010); Lai et al. (2015a, 2013, 2012); Linden (2008); Mo et al. (2014b); Barrow et al. (2009); Rouse and Krueger (2004). Anecdotal evidence suggests that pedagogy-focused CAL

test score gains of 0.47σ in two years from a CAL program that allowed some personalization. Our results finding large test-score gains for voucher winners, combined with question-level data showing how the Mindspark software addressed the considerable heterogeneity in student preparation are consistent with this idea, and suggest that personalization may be a key ingredient for achieving the full potential of technology-aided instruction.

Second, we contribute three key facts to the literature on education in developing countries. Prior work has posited with indirect evidence that a likely reason for low education productivity in these settings may be the combination of curricula designed for high-achieving students, increasing heterogeneity in student preparation resulting from rapid expansion of education access, and curricular instruction therefore being at a level and pace that may be too high for most students (Glewwe et al., 2009; Banerjee and Duflo, 2012; Pritchett and Beatty, 2015). We provide direct evidence in support of this hypothesis by documenting (a) large gaps between student preparation and grade-level standards that grow by grade, (b) considerable heterogeneity in student preparation in the same grade, and (c) a lack of progress in learning under the status quo for students with low initial learning levels.

Third, we contribute to the evidence on policy options to address the challenge of variation in student preparation. The most promising approaches to date have involved either explicit tracking of early grade classrooms (Duflo et al., 2011) or grouping primary school students by their level of preparation and teaching them basic skills (Banerjee et al., 2007, 2016). However, it is not clear if this approach can be extended to secondary grades where the content is more advanced and complex, and the heterogeneity in student preparation is much higher (exacerbated by “automatic grade promotion” policies in many countries). These conditions of large variation in within-classroom student preparation make the effective delivery of *any* curriculum challenging even for highly motivated and trained teachers (in both developed and developing countries), and our results suggest that personalized technology-aided instruction may be a promising option for improving student learning in such settings.

Finally, our results speak more broadly to the potential for new technologies to enable low-income countries to leapfrog constraints to development. Examples from other sectors include the use of mobile telephones to circumvent the lack of formal banking systems (Jack and Suri, 2014), the use of electronic voting machines for better enfranchisement of illiterate citizens in democracies (Fujiwara, 2015) and the use of biometric authentication to circumvent literacy constraints to financial inclusion (Muralidharan et al., 2016). Our results point to the possibility that technology-aided instruction could help accelerate the improvement of education quality in developing countries, by enabling them to circumvent constraints to human capital creation in low-income settings such as low teacher and parent human capital.

interventions have typically focused on grade-appropriate content in response to schools’ and teachers’ preference for CAL software to map into the topics being covered in class and reinforce them.

The rest of this paper is organized as follows. Section 2 describes the intervention, and experimental design. Section 3 describes our data. Section 4 presents our main results. Section 5 discusses cost-effectiveness, and policy implications. Section 6 concludes.

2 Intervention and Study Design

2.1 The Mindspark CAL software

The Mindspark CAL software, developed by Educational Initiatives (a leading Indian education firm), is the central component of the program we study. The software is interactive and includes continuous student assessment alongside instructional games, videos, and activities from which students learn through explanations and feedback. It reflects over a decade of product development and aims to leverage several posited channels by which education technology may improve pedagogy. We highlight some of the key design features of the software here, and provide a more detailed description with examples for each of the points below in Appendix C.

First, it is based on an extensive corpus of *high-quality instructional materials*, featuring an item bank of over 45,000 test questions, iterated over several years of design and field testing. The design of the content tries to reflect current research in effective pedagogy that is relevant to low-income settings, such as the use of same-language subtitling for teaching literacy (Kothari et al., 2002). Further, the software allows this material to be *delivered with uniform consistency* to individual students, thereby circumventing both limitations in teacher knowledge as well as heterogeneity in knowledge and teaching ability across teachers.

Second, the content is *adaptive*, with activities presented to each student being based on that student’s performance. This adaptation is dynamic, occurring both at the beginning based on a diagnostic assessment, and then with every subsequent activity completed. Thus, while the Mindspark content database is mapped into the grade-level curricular standards of the education system, an essential feature of the software is that the content presented to students is not linked to the curriculum or textbook of the grade that the student is enrolled in. In other words, it enables dynamic “Teaching at the right level” for each individual student and can cater effectively to very wide heterogeneity in student learning levels that may be difficult for even highly trained and motivated teachers to achieve in a classroom setting.

Third, even students at approximately similar levels of understanding of a topic, may have very different specific areas of conceptual misunderstanding. Thus, the pedagogical approach needed to alleviate a student-specific conceptual ‘bottleneck’ may be different across students. Mindspark aims to address this issue by using its large database of millions of student-question level observations to identify patterns of student errors and to classify the type of error and target *differentiated remedial instruction* accordingly (see Appendix C.4.2 for examples). This

attention to understanding patterns in student errors builds on an extensive literature in education that emphasizes the diagnostic value of error analysis in revealing the heterogeneous needs of individual students (see Radatz 1979 for a discussion). However, while the value of error analysis is well-known to education specialists, implementing it in practice in classroom settings is non-trivial and the use of technology sharply reduces the cost of doing so.⁶

Finally, the interactive user interface, combined with the individualization of material for each student, facilitates children’s *continuous engagement* with the material. The software makes limited use of instructional videos (where student attention may waver and cannot be monitored), choosing instead to instruct with steps that require students to constantly interact with the system. This approach aims to boost student attention and engagement, to provide feedback at the level of each intermediate step in solving a problem, and to shorten the feedback loop between students attempting a problem and learning about their errors and how to correct them.

2.1.1 The Mindspark centers intervention

The Mindspark CAL software has been deployed in various settings: private and government schools, after-school instructional centers and individual subscription-based use at home. Here, we evaluate the supplementary instruction model, delivered in stand-alone Mindspark centers that target students from low-income households. Students sign up for the program by selecting a 90-minute slot, outside of school hours, which they are scheduled to attend six days per week. The centers charged a (subsidized) fee of INR 200 (USD 3) per month.⁷

Scheduled daily instruction in Mindspark centers is divided into 45 minutes of computer-based instruction and 45 minutes of supervised instructor-led group-based study. In the time allotted to the computer-based instruction, each child is assigned to a Mindspark-equipped computer with headphones that provides him/her with activities on math, Hindi and English. Two of the days of the week are supposed to be devoted to math activities, two days to Hindi, one day to English, and one day in which the child can choose the subject.

The group-based instruction component includes all students in a given batch (typically around 15 students) and is supervised by a single instructor. Instructors are locally hired and are responsible for monitoring students when they are working on the CAL software, providing the group-based instruction, facilitating the daily operation of the centers, and

⁶The emphasis on error analysis reflects EI’s long experience in conducting similar analyses and providing diagnostic feedback to teachers based on paper-and-pen tests (Muralidharan and Sundararaman, 2010). Thus, the Mindspark development process reflects the aim of EI to use technology to improve productivity in implementing ideas that are believed by education specialists to improve the effectiveness of pedagogy.

⁷The intensity of the program, as well as the fee charged, was designed to be comparable to after-school private tutoring, typically conducted in groups of students, which is common in India. According to the 2012 India Human Development Survey, 43% of 11-17 year olds attended paid private tutoring outside of school.

encouraging attendance and retention of enrolled students.⁸ Instruction in the group-based component consists of supervised homework support and review of core concepts of relative broad relevance for all children without individual customization.

The intervention is, thus, a blended learning model that includes personalized one-on-one computer-aided instruction along with additional group academic support by an instructor. As a result, all our estimates of program impact and cost effectiveness are based on this composite program. However, in section 5.2, we present several pieces of descriptive evidence suggesting that most of the gains in test scores we find are likely to be attributable to the CAL component rather than the group instruction. We also argue that our results are likely to under-estimate the full potential of a blended learning model because the Mindspark centers did not optimize the use of the instructor for pedagogy.

2.2 Sample

The intervention was administered in three Mindspark centers in Delhi focused on serving low-income neighbourhoods. The sample for the study was recruited in September 2015 from five public middle schools close to the centers. All five schools had grades 6-8, three of these schools had grade 9, and only two had grades 4-5. Three were girls-only schools and the other two were boys-only secondary schools. Therefore, our study sample has a larger share of girls in grades 6-8. In each school, with authorization from the school principals, staff from EI and from J-PAL South Asia visited classrooms from grades 4-9 to introduce students to the Mindspark centers intervention and the study and to invite them and their parents to a scheduled demonstration at the nearby center. Students were provided flyers to retain this information and to communicate with their parents.

At the demonstration sessions, students and their parents were introduced to the intervention by staff from EI and basic background information was collected. Parents were told that, if their child wanted to participate in the study, he/she would need to complete a baseline assessment and that about half of the students would be chosen by lottery to receive a voucher which would waive the usual tuition fees of INR 200 per month until February 2016 (i.e. for nearly half of the school year). Students who were not chosen by lottery were told that would be provided free access to the centers after February 2016, if they participated in an endline assessment in February 2016. However, lottery losers were not allowed to access the program during the study period. These two design features helped to reduce attrition, and increase statistical power respectively.

⁸These instructors are recruited based on two main criteria: (a) their potential to interact with children; and (b) their performance on a very basic test of math and language. However, they are not required to have completed a minimum level of education at the secondary or college level, or have any teacher training credentials. They receive an initial training, regular refresher courses, and have access to a library of guiding documents and videos. They are paid much lower salaries than civil-service public-school teachers.

Our study sample comprises the 619 students who completed the baseline tests and surveys. About 97.5% of these students were enrolled in grades 6-9.⁹ To assess the representativeness of our self-selected study sample (and potential limitations to the external validity of our results), we compare the test score mean and distribution of study participants, and that of the full population of students in the same schools using administrative data on final exam scores in the preceding school year (2014-15). While study participants have modestly higher pre-program test scores (of around 0.15σ) than non-participants (Table A.1), there is substantial common support in the range of achievement across participants and non-participants (Figure A.1), suggesting that our results are likely to extend to other students in this setting.

2.3 Randomization and Compliance

The 619 participants were individually randomized into treatment and control groups with 305 students in the control and 314 in the treatment group. Randomization was stratified by center-batch preferences.¹⁰ The treatment and control groups did not differ significantly on any observable dimension at baseline (Table 1, Panel A). Of the 314 students offered a voucher for the program, the mean attendance rate was 58% (around 50 days out of a maximum possible of 86 days). The full distribution of attendance among lottery-winners is presented in Figure A.3, and we present both ITT estimates of winning the lottery and IV estimates of the dose-response relationship as a function of days of attendance in Section 4.

Of the 619 students who participated in the baseline test, 533 also attended the endline test (270 control and 263 treatment students). The follow-up rate was 84% in the treatment group, and 88% in the control group, but the difference was not significant at the 5% level. We also find no significant difference between treatment and control groups in mean student characteristics (age, gender, SES, or baseline test scores) of those who attend both the baseline and endline test, and comprise our main study sample (Table 1, Panel B).

3 Data

3.1 Student achievement

The primary outcome of interest for this study is student test scores. Test scores were measured using paper-and-pen tests in math and Hindi prior to the randomization (September 2015, baseline) and near the end of the school year (February 2016, endline).¹¹ Tests were

⁹589 students were enrolled in grades 6-9, 15 were enrolled in grades 4-5 and, for 15 students, the enrolled grade was not reported. Our focus on Grades 6-9 reflects our funding from the JPAL Post Primary Education Initiative, which prioritized studying interventions to improve post-primary education (after fifth grade).

¹⁰Students were asked to provide their preferred slots for attending Mindspark centers given school timings and other commitments. Since demand for some slots was expectedly higher than others, we generated the highest feasible slot for each student with an aim to ensure that as many students were allocated to their first or second preference slots as possible. Randomization was then carried out within center-by-batch strata.

¹¹It was important to test students in a pen-and-paper format, rather than computerized testing, to avoid conflating true test score gains with greater familiarization with computer technology in the treatment group.

administered centrally in Mindspark centers at a common time for treatment and control students with monitoring by J-PAL staff to ensure the integrity of the assessments.

The tests were designed independently by the research team and intended to capture a wide range of student achievement. Test items ranged in difficulty from “very easy” questions designed to capture primary school level competencies much below grade-level to “grade-appropriate” competencies found in international assessments. Test scores were generated using Item Response Theory models to place all students on a common scale across the different grades and across baseline and endline assessments. The common scale over time allows us to characterize the *absolute* test score gains made by the control group between the two rounds of testing. The assessments performed well in capturing a wide range of achievement with very few students subject to ceiling or floor effects. Details of the test design, scoring, and psychometric properties of individual test questions are provided in Appendix D.

3.2 Mindspark CAL system data

The Mindspark CAL system collects detailed logs of all interactions that each student has with the software platform. This includes, for example, daily attendance, the estimated grade level of student achievement as determined by the Mindspark system, the record of each question that was presented to the child and whether he/she answered correctly, as well as other details such as time taken to answer or enter keystrokes to measure engagement with content. These data are available (only) for the treatment group for the duration of the intervention. We use these data in three ways: to describe the mean and distribution of learning gaps relative to curricular standards in each grade at baseline; to demonstrate the personalization of instruction at the core of the Mindspark system; and to characterize the evolution of student achievement in the treatment group over the period of the treatment.

3.3 School records

At the school level, we collected administrative records on test scores on school exams of all study students and their peers in the same schools and classrooms. This was collected for both the 2014-15 school year (to compare the self-selected study sample with the full population of students in the same schools) and the 2015-16 school year (to evaluate whether the treatment affected test scores on school exams).

3.4 Student data

At the time of the baseline assessment, students answered a self-administered written student survey which collected basic details about their socio-economic status, and household characteristics. A shorter survey of time-varying characteristics was also administered at endline. Additionally, we also conducted a brief telephone survey of parents in July 2016 to collect information on use of private tutoring during the prior academic year, and their opinion of the Mindspark program.

4 Results

4.1 Business-as-usual academic progress

Our first set of results use the rich CAL data on student achievement to present a more granular characterization of the variation in learning levels within a typical classroom than has been possible so far. Specifically, the data from the Mindspark CAL system provides an assessment of the actual grade-level of each student’s learning level regardless of grade enrolled in. The main results are presented in Figure 1, which shows the full joint distribution of the grades students were enrolled in and their assessed learning level at the start of treatment.¹²

We highlight three main patterns in Figure 1. First, most children are already much below grade level competence at the beginning of post-primary education. In grade 6, the average student is about 2.5 grades behind in math and about half a grade behind in Hindi.¹³ Second, although average student achievement is higher in later grades, indicating some learning over time, the slope of achievement gains (measured by the line of best fit) is considerably flatter than the line of equality between curricular standards and actual achievement levels. This suggests that average student academic achievement is progressing at a lower rate than envisaged by the curriculum — by grade 9, students are (on average) nearly 4.5 grades behind in math and 2.5 grades behind in Hindi. Third, the figure presents a stark illustration of the very wide dispersion in achievement among students enrolled in the *same* grade: students in our sample span 5-6 grade levels in each grade. Characterizing this variation is essential for designing effective instructional strategies, and the CAL data allow us to do this more clearly than the literature to date, by precisely characterizing the learning level of *each* student.

Our second descriptive result uses the IRT-linked student-level panel data from our independent tests to investigate heterogeneity in the ‘business as usual’ academic progress in the control group. Specifically, we present the value-added of test scores for the bottom, middle, and top third of the within-grade achievement distributions in our sample in both math and Hindi. We estimate the following regression (without a constant term):

$$Y_{is2} = \gamma \cdot Y_{is1} + \alpha_j \cdot \text{terc}_{js1} + \epsilon_{is2} \quad (1)$$

¹²Note that these data are only available for students in the treatment group. However, Figure 1 uses data from the *initial* diagnostic test, and does not reflect any instruction provided by Mindspark.

¹³These figures are likely to underestimate the gap relative to curricular standards at the end of primary school (grade 5), because weaker students are less likely to have made the transition to post-primary school enrolment and will not be in our sample. Also, the allocation of test questions to grade levels is more robust in math than language (where competencies are less well-delineated across grades). Thus, although most patterns across grades are similar in the two subjects, the computer system’s assessment on grade-level competence of children is likely to be more reliable for math. Baseline test scores on our independent tests in both subjects are consistently higher for students assessed by the CAL program as being at a higher grade level of achievement, which helps to validate the grade-level bench-marking by the CAL program. Further details of the diagnostic test and bench-marking by the software are presented in Appendix C.

where Y_{ist} is student i 's test score on our independent assessment in subject s at period t (normalized to $\mu=0$, $\sigma=1$ on the baseline test), $\mathbf{terc}_{js1}$ is a vector of indicator variables for the within-grade terciles (j) of baseline score in subject s , and ϵ_{is2} is the error term.

Coefficients from the vector α_j represent the absolute value-added in each tercile j . Such an interpretation is made feasible by the comparable measurement of test scores across baseline and endline tests on a common scale, which is done here by linking test scores using IRT. These coefficients are presented in Figure 2, and we see considerable heterogeneity in student progress by initial learning level. While students in the top third of the baseline test-score distribution show significant academic progress between baseline and endline, it is striking that we cannot reject the null of *no increase* in test scores for the bottom-third of students in the control group over the same period (with coefficients close to zero in both subjects).

Figures 1 and 2 highlight that a vast majority of students in this setting are left behind the curriculum, and that this gap grows by grade. Further, lower-performing students appear to make no academic progress at all. To the best of our knowledge, we are the first to present direct evidence on these patterns in developing countries. However, they are consistent with evidence on slow test score growth within cohorts over time in repeated cross-sections in India (Pritchett, 2013), and with the patterns of results observed in experimental evaluations of education interventions in developing countries in the past decade (Glewwe et al., 2009; Duflo et al., 2011; Banerjee and Duflo, 2012). Thus, although our sample is not representative and we present these results mainly to set the context for the intervention, the patterns presented here are likely to be similar to those in other developing country settings.

4.2 Program Effects (Intent-to-treat estimates)

The main treatment effects can be seen visually in Figure 3, which presents mean test scores in the baseline and endline assessments in math and Hindi for lottery-winners and losers. While test scores improve over time for both groups, endline test scores are significantly and substantially higher for the treatment group in both subjects.

We estimate intent-to-treat (ITT) effects of winning the lottery (β) using:

$$Y_{is2} = \alpha + \gamma.Y_{is1} + \beta.Treatment_i + \phi + \epsilon_{is2} \quad (2)$$

where Y_{ist} is student i 's test score in subject s at period t (normalized to $\mu=0$, $\sigma=1$ on the baseline test); $Treatment$ is an indicator variable for being a lottery-winner; ϕ is a vector of stratum dummy variables to reflect the randomization design; and ϵ_{it} is the error term.¹⁴

¹⁴We use robust Huber-White standard errors throughout the paper rather than clustered standard errors because of the individual (as opposed to group) randomization of students to treatment status. Further, common shocks from test day and venue effects are netted out by the inclusion of strata fixed effects since all students in the same stratum (both treatment and control), were tested on the same day in the same location.

We find that students who won the lottery to attend Mindspark centers scored 0.36σ higher in math and 0.22σ higher in Hindi compared to lottery losers after just 4.5 months (Table 2: Cols. 1-2). In Cols. 3 and 4, we omit strata fixed effects from the regression, noting that the constant term α in this case provides an estimate of the absolute value-added (VA) in the control group over the course of the treatment.¹⁵ Expressing the VA in the treatment group ($\alpha+\beta$) as a multiple of the control group VA (α), our results indicate that lottery-winners made twice the progress in math, and 2.5 times the progress in Hindi, compared to lottery-losers. These are ITT results based on an average attendance of about 58% among lottery-winners. We present IV results and estimates of a dose-response relationship in Section 4.6.

In addition to presenting impacts on a normalized summary statistic of student learning, we also present impacts on the fraction of questions answered correctly on different domains of subject-level competencies (Table 3). The ITT effects are positive and significant across all domains of test questions. In math, these range from a 12% increase on the easiest type of questions (arithmetic computation) to a 36% increase on harder competencies such as geometry and measurement. Similarly, in Hindi, ITT effects range from a 7% gain on the easiest items (sentence completion) to a 19% gain on the hardest competence (answering questions based on interpreting and integrating ideas and information from a passage).

4.3 Heterogeneity

We investigate whether ITT effects vary by gender, socio-economic status, or initial test scores, using a linear interaction specification and find no evidence of heterogeneity on these dimensions (Table 4). Since baseline test scores are a good summary statistic of prior inputs into education, we also present non-parametric estimates of the ITT effect as a function of baseline scores. We do this by plotting kernel-weighted locally-smoothed means of the endline test scores at each percentile of the baseline test-score distribution, separately for the treatment and control groups (Figure 4). In both math and Hindi, we see that the test scores in the treatment group are higher than those in the control group at every percentile of baseline test scores, and that the gains appear similar at all percentiles.

Next, we formally test for equality of treatment effects at different points of the within-grade test-score distribution. We do this by regressing endline test scores on the baseline test scores, indicator variables for treatment and for within-grade terciles at baseline, and interaction terms between the treatment variable and two terciles (the regression is estimated without a constant). We see no evidence of heterogeneity here as well (Table 5). The coefficient on the

¹⁵Again, note that this interpretation is possible because the baseline and endline tests are linked to a common metric using Item Response Theory: α in Eq. 2 is the mean of the (α_j) coefficients estimated in Eq. 1 and shown in Figure 2. This would not be possible if scores were normalized within grade/period as is common practice. However, treatment effects are very similar (0.36σ in math and 0.21σ in Hindi) when test scores are normalized relative to the within-grade distribution in the control group at the endline (Table A.2).

treatment dummy itself is statistically significant, but the interaction terms of treatment with the tercile at baseline are statistically indistinguishable from zero.

Thus, winning a voucher appears to have benefited students at all parts of the achievement distribution significantly and relatively equally, suggesting that the Mindspark software could teach all students equally well. Recall, however, that students in the lowest tercile of the within-grade baseline test score distribution did not make any academic progress in the control group on either subject (Figure 2; Table 5: Row 1). Thus, while the *absolute* test score gain from winning a Mindspark voucher was similar for all lottery winners, the *relative* gains from the treatment (measured as a multiple of what students would have learnt in the absence of treatment) was much larger for the weaker-performing students.

4.4 Personalization

These large test score gains reflect the composite effects of multiple features of Mindspark (as described in Section 2), and our experiment was not designed to separately identify the effects of these channels. However, the detailed question-level data available in the system for individual students in the treatment group allow us to closely examine a potentially important component for the program’s effectiveness: the delivery of personalized instruction which is able to target student preparedness precisely and update instruction appropriately.

We first present direct evidence that the software precisely targeted instructional material at an individual student’s level. Figure 5 shows, separately by each grade of school enrolment, the actual grade level of students’ academic preparedness as estimated by the Mindspark CAL system and the grade-level difficulty of the questions that students were presented.¹⁶ Across the horizontal axis on each subgraph, we see the wide dispersion in academic preparedness *within each grade*, reiterating our interpretation of Figure 1. On the vertical axis, we see that the software was able to precisely target instruction to student preparedness and that the typical student is presented items, either at their grade level or at adjacent grade levels. Thus, the software was able to simultaneously accommodate variation in student preparation spanning five to grade levels for students enrolled in the same grade. This degree of personalization is considerably more precise than would be feasible for a single teacher to deliver to all students in a standard classroom setting.

Second, we show that the CAL system accommodates variation, not just in the incoming levels of student preparedness, but also in their pace of learning and individual trends in student achievement. Figure 6 presents non-parametric plots of the average difficulty level of the math items presented to students over the course of the intervention, documenting

¹⁶In both math and Hindi, we use data from a single day which is near the beginning of the intervention, after all students would have completed their initial assessment, and when Mindspark computer-aided instruction in the relevant subject was scheduled in all three centers.

that the software updates its estimate of student achievement levels in real time and modifies instruction accordingly.¹⁷ In the first figure, separate lines are plotted by the grade students are enrolled in and, in the second figure, by their initial level of achievement. As can be seen, this estimated level of difficulty increases for all groups of students continuously indicating that students were making regular progress during the study period, and that the software was able to customize instruction to their increasing achievement. The individualization of the dynamic updating of content is highlighted further in Figure A.2 where we use the rich individual data on student achievement to plot similar trajectories separately for *each student* in the treatment group.

In summary, Mindspark seems to be able to deliver customized instruction at a level that is unlikely to be feasible under most current models of classroom-based instruction. This personalization is likely to be especially important for academically weaker students whose preparation is several grade levels behind their enrolled grade, and who (likely as a result) appear to make no progress under regular classroom-based instruction as seen earlier. Note that the CAL interventions reviewed in Appendix B—with the exception of Banerjee et al. (2007)—typically featured grade-appropriate instruction. These grade-linked CAL programs find only modest effects despite combining most other advantages of CAL such as high-quality content delivered in a standardized way, interactive content to promote student engagement, and immediate feedback. We conjecture therefore that a likely reason for the large effects we find is the ability of the software to finely personalize instruction and “Teach at the Right Level” to all students, especially in this context of widespread learning deficits.¹⁸

4.5 Grade-level impact decomposition, and impacts on school tests

One consequence of the substantial deficits in student preparation (Figure 1), is that even large absolute increases in learning may not be sufficient for raising test scores on grade-level assessments since the gains in learning could be taking place below grade-level. We therefore use the CAL system data to directly examine the grade-level distribution of content presented by the software to students in the treatment group (see Figure 7). Consistent with the pattern of improvement in math test scores over time seen in the left panel of Figure 6 (which take place far below the level of the enrolled grade), we see that in math, most of the content presented to students by Mindspark was below grade level, with very little content at the level

¹⁷We study dynamic updating only in math because the algorithms for updating are more finely developed, and differ importantly from Hindi. In math, the software moves students to a harder question in the same competence after answering initial question(s) correctly but in Hindi the software ensures that a student has mastered all competencies at a grade level before being presented questions at the next grade level.

¹⁸In future work, we hope to work with EI to run micro-experiments on the Mindspark platform to try to isolate the impact of personalization on learning outcomes. However, from the perspective of improving productivity in the delivery of education, we are more interested in the composite “full potential” impact of technology-aided learning in improving education outcomes, which is what we focus on in this paper.

of the grade the student is enrolled in. However, in Hindi, in addition to lower-grade content, a substantial portion of the Mindspark instruction in each grade was at grade level.

Next, we examine the extent to which the treatment effects on our independent tests map into the grade-level patterns of content presented by the CAL system by separately presenting treatment effects on the proportion of test questions answered correctly at and at below grade level.¹⁹ We present these results in Table 6 and see that while there were large treatment effects in math on items below grade level, there was no impact on grade-level questions. In Hindi, on the other hand, we find that the treatment effect is significant for both questions at and below grade level.

These patterns in our data are also replicated in the independent data we collected on test scores on school exams. Table 7 presents the treatment effect of being offered a voucher on scores on the annual end of year school exams held in March 2016.²⁰ Mirroring the results on grade-level items on our own tests, we find a significant increase in test scores of 0.19σ in Hindi but no significant effect on math. We also do not find significant effects on the other subjects (science, social science, or English), although all the point estimates are positive.

Both sets of results (on our tests and on the school tests) are consistent with the data on grade-level distribution of questions presented by Mindspark shown in Figure 7, and highlight the importance of careful test construction for education research in settings with a wide distribution of student learning levels. In particular, relying on grade-level assessments would have led to erroneous inference regarding program impacts (especially in math). See Appendix D for further details on test design for our study, and Muralidharan (2017) for a detailed discussion on test construction for education research in general.

4.6 IV estimates of dose-response relationship

All the results presented so far are ITT estimates, which are based on an average attendance of about 58% among lottery-winners.²¹ In this section, we present LATE estimates of the impact of actually attending the Mindspark centers, and (with further assumptions) estimates of predicted treatment effects at different levels of program exposure. We estimate the dose-response relationship between days of attendance and value-added using:

$$Y_{is2} = \alpha + \gamma.Y_{is1} + \mu_1.Attendance_i + \eta_{is2} \quad (3)$$

¹⁹Items on our tests, which were designed to capture a wide range of achievement, were mapped into grade-levels with the help of a curriculum expert.

²⁰In Delhi, test papers for the annual exam are common across schools for each subject in each grade. In our regressions, we normalize test scores to $\mu=0$, $\sigma=1$ in each grade/subject in the control group.

²¹About 13% of lottery-winners attended for one day or less. The mean attendance among the rest was 57 days (around 66%). Figure A.3 plots the distribution of attendance among lottery winners, and Table A.3 presents correlations of attendance among lottery winners with various baseline characteristics.

where Y_{ist} is defined as previously, *Attendance* is the number of days a student logged in to the Mindspark system (which is zero for all lottery-losers) and η_{ist} is the error term. Since program attendance may be endogenous to expected gains from the program, we instrument for *Attendance* with the randomized offer of a voucher.

The IV estimates suggest that, on average, an extra day of attending the Mindspark centers increased test scores by 0.0065σ in math and 0.004σ in Hindi (Table 8: Cols. 1-2). These estimates identify the average causal response (ACR) of the treatment which “captures a weighted average of causal responses to a unit change in treatment (in this case, an extra day of attendance), for those whose treatment status is affected by the instrument” (Angrist and Imbens, 1995). Using these IV estimates to predict the effect of varying the number of days attended requires further assumptions about (a) the nature of heterogeneity in treatment effects across students (since the ACR is only identified over a subset of compliers, and not the full sample) and (b) the functional form of the relationship between days attended and the treatment effect (since the ACR averages causal effects over different intensities of treatment).

We present three pieces of suggestive evidence that constant treatment effects across students may be a reasonable assumption in this setting. First, the ITT effects were constant across the full distribution of initial achievement, which is a good summary measure for relevant individual-specific heterogeneity (Figure 8, Tables 3, 4). We also found no significant evidence of treatment heterogeneity across observed pre-treatment characteristics (Table 3).

Second, we cannot reject the equality of the IV estimates of Eq.(3) and the OLS estimates using a value-added (VA) specification (Table 8: Cols. 3-4), which suggests that the ATE and the LATE may be similar here. For both math and Hindi, the p-value from the difference-in-Sargan test (similar to a Hausman test, but allowing for heteroskedasticity) testing equivalence of OLS and IV results is substantially greater than 0.1 (Cols. 1-2).²²

Finally, the constant term in the OLS VA specifications (corresponding to zero attendance) is identical when estimated using the full sample and when estimated using only the data in the treatment group (Table 8: Cols. 3-6). The constant term is identified using both the control group and “never-takers” when using the full sample, but is identified over only the “never-takers” when the sample is restricted to lottery-winners. Thus, the outcomes for the “never takers” are similar to those for the control group, suggesting equality of potential outcomes across different compliance groups.²³

²²Note that this close correspondence between the OLS VA and IV estimates is consistent with much recent evidence that VA models typically agree closely with experimental and quasi-experimental estimates (see, for instance Chetty et al. (2014); Deming et al. (2014); Singh (2015); Angrist et al. (2016))

²³This test is similar in spirit to tests suggested by Bertanha and Imbens (2014) and Brinch et al. (2016), for extending the validity of RD and IV estimates beyond LATE to average treatment effects.

We next explore the functional form of the relationship between days attended and the treatment effect directly by plotting value-added against attendance using data for the lottery winners (Figure 8). The relationship seems entirely linear in math and, even in Hindi, although there are some signs of diminishing returns, we cannot reject a linear dose-response relationship (see Table A.4). A linear dose-response is additionally plausible when considering the adaptive nature of the intervention which allows it to be equally effective regardless of the starting point of the student (thus alleviating possible concerns of diminishing returns to treatment, at least over the relatively short duration of treatment in this study).

Under the assumptions of constant treatment effects and a linear dose-response relationship, both of which appear reasonable in this context, our IV results suggest that attending Mindspark centers for 90 days, which roughly corresponds to half a school year with 80% attendance, would lead to gains of 0.59σ in math and 0.37σ in Hindi (last row of Table 8). We extrapolate results to 90 days, rather than a full school year, to keep the predictions near the range of the program exposure provided by our experiment (the maximum was 86 days).

These estimates are conservative and likely to understate the dose-response relationship because the *Attendance* variable includes time spent in the Mindspark centers on instruction in other subjects that we do not test (especially English).²⁴ In Table A.5, we present analogous IV and value-added estimates which only account for days spent by students on the subjects that we test (math and Hindi). Using these results, and the same assumptions as above, we estimate that 90 days of Mindspark attendance, split equally between the two subjects, would lead to test score gains of 0.76σ in math and 0.5σ in Hindi.

4.7 Robustness

4.7.1 Attrition

Since the difference in attrition between the treatment and control groups is significant at the 10% level (Table 1), we test the robustness of our results to attrition by modeling selection into the endline based on observed characteristics, and present inverse probability weighted treatment effects: the estimated ITT effects are substantively unchanged (Table A.6). We also compute Lee (2009) bounds for the ITT effect: although bounds are wide, the treatment effects are always positive and significant (Table A.7).

4.7.2 Familiarity with test questions

Our independent tests used items from several external assessments, some of which (in the Indian setting) were designed by EI; this raises the possibility that results on our assessments are overstated due to duplication of items between our tests and the Mindspark

²⁴See Muralidharan and Sundararaman (2015) for an illustration of the importance of accounting for patterns of time use across subjects for inference regarding the productivity of education interventions.

item bank. Note that this item bank contains over 45,000 items and so mere duplication in the database does not imply that a student would have been presented the same item during the intervention. Nevertheless, we test for this concern by computing the treatment effect expressed as the proportion correct on items from EI assessments and items from other assessments. The ITT effects are positive, statistically significant and of similar magnitude for both sets of items in math and Hindi (Table A.8).

4.7.3 Private Tutoring

Our results may also be confounded if winning a Mindspark voucher led to changes in the use of private tutoring. To test for this possibility, we collected data from parents of study students, using phone surveys, on whether the student attended paid extra tutoring (other than Mindspark) in any subject for each month from July 2015 to March 2016. Dividing this period into “pre-intervention” (July to September 2015) and “post-intervention” (October 2015 to March 2016), we test whether winning a Mindspark-voucher affected the incidence of private tutoring in the “post-intervention” period. We present these results in Table A.9. While there is a modest increase in private tutoring for all students in the post-treatment period (consistent with increased tutoring closer to annual school exams), we find no evidence of any differential use of private tutoring among lottery winners.

5 Cost-Effectiveness and Discussion

5.1 Cost-effectiveness

Since we evaluate an after-school program, our treatment effects reflect both the productivity of the Mindspark centers and the effects of additional instructional time. Thus, a natural comparison of cost effectiveness is with after-school private tutoring, which is widespread in our setting. A nearly direct comparison is made possible by a contemporaneous study conducted in the same location and student age group: Berry and Mukherjee (2016) report results from a randomized evaluation that studied the impact of paid private tutoring on learning outcomes of middle-school students (in grades 6-8) in Delhi at the same time as our study. The program also provided six days of instruction per week, for two hours per day (versus 90 minutes per day at Mindspark centers), and also charged INR 200 per month. The private tutoring program was run by a well-respected and motivated non-profit organization, Pratham, who have run several education programs in India that have been found to have significant positive impacts on student learning (see, for example, Banerjee et al. (2007, 2016)).

Despite several similarities, this after-school tutoring program differed from the Mindspark centers in two important ways: First, instruction focused on reinforcing grade-level curriculum and was not customized to students’ academic preparation; second, the instruction was delivered in person by a tutor in groups of up to 20 students (a similar ratio of instructor to

student as seen in Mindspark centers), but did not make use of any technology for instruction. Both these features are typical of the existing private tutoring market in India. At the end of a year of the program, Berry and Mukherjee (2016) find no impact on student test scores in independent assessments of either math or English despite the program having spent more than twice the after-school instructional time provided by the Mindspark centers during our evaluation (33% more instruction/week, and evaluated after a full year as opposed to 4.5 months). These results suggest that additional after-school instruction by itself, may not have been enough to improve learning outcomes in this setting.²⁵

A second policy-relevant comparison is with the productivity of government-run schools (from where the study subjects were recruited). Per-pupil monthly spending in these schools in Delhi was around INR 1500 (USD 22) in 2014-15; students spend 240 minutes per week on math and Hindi; and we estimate that the upper-bound of the value-added in these schools was 0.36σ in math and 0.15σ in Hindi over the 4.5-month study period. Specifically, this was the *total* value-added in the control group in Table 2, which also includes the effects of home inputs and private tutoring, and therefore likely over-estimates the value-added in public schools.

Using our ITT estimates, we see that Mindspark added 0.36σ in math and 0.22σ in Hindi over the same period in around 180 minutes per week on each subject. The Mindspark program, as delivered, had an unsubsidized cost of about INR 1000 per student (USD 15) per month. This includes the costs of infrastructure, hardware, staffing, and also the pro-rated costs for software development. Thus, even when implemented with high fixed costs and without economies of scale, and based on 58% attendance, providing access to the Mindspark centers delivered greater learning at lower financial and time cost than default public spending.

Steady-state costs of Mindspark at policy-relevant scales are likely to be much lower since the (high) fixed costs of product development have already been incurred. If implemented in government schools, at even a modest scale of 100 schools, per-pupil costs reduce to about USD 2 per month (including hardware costs but excluding rent and utilities); at a scale of 1000 schools, these reduce to less than USD 1 per month. At the scale of an Indian state, the per-pupil variable costs (software maintenance and technical support) are expected to be below USD 2 annually, which is a small fraction of the USD 150 annual cost (over 10 months) during our pilot. The program thus has the potential to be very cost-effective at scale.

Finally, while education spending can increase continuously over time, student time is finite, which makes it interesting to evaluate the effectiveness of education interventions per unit of time (independent of financial cost). A useful point of comparison is provided by Muralidharan (2012), who finds that providing individual-level performance bonuses to teachers in India

²⁵While Berry and Mukherjee (2016) do not identify specific reasons for their results, the patterns we present in Figures 1 and 2 suggest that one plausible reason may be that additional time spent on instruction at a grade-appropriate level may not yield much learning for students who are several grade levels behind.

led to test score gains of 0.54σ and 0.35σ in math and language for students exposed to the program for five years. This is one of the largest effect sizes seen to date in an experimental study on education in developing countries. Yet, we estimate that regularly attending Mindspark centers for half a year would yield similar gains (in one tenth the time).

Figure 6 suggests that the ITT effects on math were close to a full grade-level over just 4.5 months (even with only 58% attendance). Thus, using Mindspark regularly in schools may be an especially promising option for helping to bridge the large gaps in student readiness within time frames that may make it feasible for lagging students to catch up to grade-level standards of instruction. Testing this possibility is an important topic for future research.

5.2 Interpreting a composite treatment effect

Each session at a Mindspark center combines 45 minutes of computer-led individual instruction, and 45 minutes of group teaching by an instructor. Given the composite nature of the intervention, we cannot experimentally decompose the relative contribution of these two components. Our estimates should therefore be interpreted as those of a “blended learning” program, and all our cost effectiveness calculations are based on the composite program.

Yet, there are several good reasons to believe that most of the impacts we find are likely to be attributable to the CAL component as opposed to the group instruction. First, the direct comparison with the results in Berry and Mukherjee (2016) suggest that group-based instruction on grade-level materials had no impact on learning in the same context even with over double the duration of exposure relative to the program we study. Second, the precise customization in the computer-based instruction that we document in Figure 5 is unlikely to be achievable for even motivated teachers when faced with groups of students with wide variation in student achievement. Third, prior evidence on positive impacts of group-based instruction has highlighted the importance of *homogenizing* the groups by learning level for effective instruction (Banerjee et al., 2007, 2016). However, in our setting, the students in a given Mindspark batch (who would attend group instruction together) had even *wider* variation in preparation than the students in our study. This is because our study focused on a post-primary sample of grades 6-9, while the centers enrolled students across grades 1-9 in the same batch, based on student timing convenience. This large variation across enrolled grade and student preparation in each batch can be seen in Figure A.4), and further suggests that the group-based instruction may have had only limited direct impact on learning.

According to EI, the main roles of the instructor were to encourage regular student attendance at the centers, to ensure time on task while in front of the computer, and to supervise homework completion and examination preparation during the group-instruction period (which parents demanded). As a result, group instruction in the centers focused mostly on homework support or the revision of materials for exams but was not optimized for

instruction. Our results are therefore likely to underestimate the full potential of well-designed blended-learning interventions (along the lines discussed further in the conclusion).

5.3 Policy Implications

Our results are directly relevant to several current education policy issues around the world. First, there is considerable interest among policy makers in using technology in education. In practice, policy makers (especially in developing countries) have mainly concentrated on policies that provide computer hardware with very little attention paid to using technology for improving pedagogy.²⁶ However, as seen in Appendix B, the evidence to date suggests that such a hardware-focused approach is unlikely to yield much improvement in learning outcomes. Our results suggest that these hardware investments are likely to yield much greater returns in terms of improved learning outcomes if attention is also paid to deploying Mindspark (or similar) software to improve pedagogy in public schools. The marginal costs of doing so are likely to be low (especially in settings where hardware exists), and such a deployment would offer a natural opportunity to test impacts at a larger scale. More generally, our results highlight the potential for improving the productivity of education spending by reallocating budgets from less to more cost effective policies (Glewwe and Muralidharan, 2016).²⁷

Second, we also speak to the broader (mis-)orientation of education systems in developing countries such as India. Scholars of the Indian education system have posited with indirect evidence that it appears to be designed more for screening high-ability students than creating human capital by teaching all students effectively (Das and Zajonc, 2010; Banerjee and Duflo, 2012; Muralidharan, 2013). We provide direct evidence in support of this hypothesis by showing that a substantial fraction of low-scoring students make no academic progress during a regular school year and show that a likely reason for this is that their learning levels are far below grade-appropriate standards. These results also point to a plausible reason for the low productivity of the Indian education system (and potentially those of other developing countries as well). Additional time and money devoted to education may not produce much learning if they are not alleviating the binding constraint of students being too far behind the curriculum. Thus, better aligning curricula and standards with actual learning levels of students may be an essential reform for improving the effectiveness of the education system.

Third, there is a contentious ongoing debate across countries regarding the desirability of ‘social promotion’ or a ‘no detention’ policy in schools. Yet, while there is research on the

²⁶For instance, various state governments in India have distributed free laptops to students in recent years. Further, progress on implementing the national-level policy on technology in education is typically measured by the number of schools with computer labs.

²⁷For instance, expensive policies such as unconditional doubling of teacher salaries in Indonesia, or providing a personal laptop to every child in Peru were promoted by advocates as effective ways of improving student learning, but were found to have no impact on learning outcomes (de Ree et al., 2015; Beuermann et al., 2015). On the other hand, allocating resources to programs like Mindspark (whose marginal cost at a similar scale are much lower) may yield considerable test-score gains at lower costs.

costs and benefits of grade retention versus automatic promotion, there is much less evidence on *how to teach effectively* in settings with high learning deficits in later grades.²⁸ One widely considered policy option is tracking by student preparation, but this may not always be feasible in practice — especially in higher grades with very wide variation in student preparation. Further, even when feasible, tracking is a controversial policy and the global evidence on its impact is mixed and inconclusive (see Betts (2011) for a review). Our results suggest that well-designed technology-led instruction can personalize instruction and enable effective teaching, even in settings of very wide within-grade dispersion in academic achievement.²⁹

The final policy-relevant point to highlight is the puzzle of low parental adoption of Mindspark despite the large positive effects we find. There is a large market for supplementary tutoring in India, and the (subsidized) Mindspark fee was set based on the market benchmarks for private tutoring. Yet, the demand for Mindspark centers was low in the absence of the (fee-waiving) vouchers and, indeed, all three centers in our study closed down soon after the conclusion of our experiment in the face of low demand (even at the subsidized price).³⁰

While studying household demand was beyond the scope of this paper, we list three possible reasons for the low demand. First, parents may not have been well-informed about the program, which may have contributed to low take up. Second, even if they take up the program, they may perceive it as ineffective if they see no improvement on school test-scores. Third, even if they see the test score gains taking place at below-grade learning levels, they may be unwilling to pay for a program that does not improve performance on high-stakes matriculation exams, which are a key pre-requisite for formal-sector jobs. These hypotheses are consistent with evidence from other settings that parents (especially low-income ones) may find it difficult to make optimal decisions regarding the appropriate investments in their child’s education (see Jensen, 2010; Dizon-Ross, 2016 for examples). More generally, while our study provides evidence regarding the ‘efficacy’ of Mindspark, the low take up illustrates the broader challenge of low adoption of effective new technologies and the need for future research on policy interventions to improve such adoption (as illustrated by Dupas, 2014).

²⁸For examples of studies evaluating the effects of grade retention, see Jacob and Lefgren (2004) in the US, Manacorda (2012) in Uruguay, and Koppensteiner (2014) in Brazil.

²⁹A further benefit of such personalization (relative to grade retention or accelerated promotion) is the ability to preserve the age-cohort based social grouping of students, which may allow for better development of socio-emotional skills, while allowing for variation in academic content presented.

³⁰The donors who subsidized the fees at Mindspark centers stipulated that they would only continue funding the subsidies if the centers could operate at or above 80% capacity (and thereby demonstrate parental willingness to pay at least the subsidized price). In practice, enrolment levels were considerably below this target, and the centers had to shut down because philanthropic funding for the subsidies ended. Since the centers shut down in March 2016, control group students who had been offered free access to the centers after the endline test, were instead offered free educational materials as compensation for participating in the study. However, Mindspark as a product is still doing well and EI continues to operate and improve the full-fee Mindspark models for higher SES families, where the demand continues to be strong.

6 Conclusion

We present experimental evidence on the impact of a technology-led supplementary instruction program in post-primary grades in urban India. We document substantial positive effects of the program on both math and language test scores and show that the program was cost-effective both in terms of time and money. The program was effective at teaching students at all levels of prior achievement, including students in the bottom-third of the within-grade distribution who are left behind by business-as-usual instruction. Using detailed information on the material presented to students in the treatment group, we demonstrate that the software was successful at targeting instruction precisely to the students' level of achievement and in handling wide variation in the academic preparation in the same grade.

These substantial effects reflect, in our opinion, the ability of the intervention to effectively target *multiple* constraints that lead to the low productivity of instructional time in Indian schools. The high quality of content, combined with effective delivery and interface, may help circumvent constraints of teacher human capital and motivation. Personalized instruction makes it possible to accommodate large deficits in initial student preparation and wide variation within a single grade. Algorithms for analyzing patterns of student errors and providing differentiated feedback and follow up content that is administered in real-time, allows for feedback that is more relevant and much more frequent. These features all reflect continuous and iterative program development over a long period of more than a decade.

These effects may plausibly be increased even further with better design. It is possible that in-school settings may have greater adherence to the program in terms of attendance. It may also be possible to improve the effectiveness of teacher-led instruction in a 'blended learning' environment by using the extensive information on student-performance to better guide teacher effort in the classroom. This 'big data' on student achievement also offers much potential of its own. In particular, such a setting may enable high-frequency randomized experiments on effective pedagogical techniques and approaches (which may vary across students) and build a stronger evidence base on effective teaching practices. This evidence may then be used to further optimize the delivery of instruction in the program and, plausibly, also for the delivery of classroom instruction. Finally, the detailed and continuous measures of effort input by the students can be used directly to reward students, with potentially large gains in student motivation, effort, and achievement.³¹

However, there are also several reasons to be cautious in extrapolating the success of the program more broadly. The intervention, as evaluated in this paper, was delivered at a modest scale of a few centers in Delhi and delivered with high fidelity on part of the providers. Such fidelity may not be possible when implementing at scale. Additional issues relate to the

³¹Direct evidence that this may be possible is provided by Hirshleifer (2015) who uses data from a (different) computer-aided instruction intervention to reward student effort and documents large effects of 0.57σ .

mode of delivery. We have only evaluated Mindspark in after-school centers and it is plausible that the effectiveness of the system may vary significantly based on whether it is implemented in-school or out-of-school; whether it is supplementary to current classroom instruction or substitutes away current instructional time; and whether it is delivered without supervision, under the supervision of current teachers or under the supervision of third parties (e.g. the Mindspark center staff). Identifying the most effective modes of delivery for the program at larger scale is an important area for future research.³²

A further point of caution is that our results should not be interpreted as supporting a de-emphasis of the role of teachers in education. Rather, since the delivery of education involves several non-routine tasks that vary as a function of individual students and situations, and requires complex contextually-aware communication, it is more likely that technology will complement rather than substitute teachers (as shown more generally by Autor et al. (2003)). So, it may be possible to improve teacher and school productivity by using technology to perform routine tasks (such as grading) and data-analysis intensive tasks (such as identifying patterns in student answers and providing differentiated feedback and instruction to students), and enabling teachers to spend more time on aspects of education where they may have a comparative advantage - such as supporting group-based learning strategies that may help build social and other non-cognitive skills that may have considerable labor market returns (Cunha et al., 2010; Heckman and Kautz, 2012; Deming, 2016).

Overall, our present study is best regarded as an efficacy trial documenting that well-designed and implemented technology-enabled learning programs can produce large gains in student test scores in a relatively short period of time. Our results suggest that the promise of technology to sharply improve productivity in the delivery of education is real, and that there may be large returns to further innovation and research on effective ways of integrating technology-aided instruction into classrooms, and on effective ways of delivering these benefits at a larger scale.

³²A useful example of such work has been the literature that followed the documenting of the efficacy of unqualified local volunteers, who were targeting instruction to students' achievement levels, in raising achievement in primary schools in two Indian cities by Banerjee et al. (2007). Subsequent studies have looked at the effectiveness of this pedagogical approach of "Teaching at the Right Level" in summer camps, in government schools and delivered alternately by school teachers and by other volunteers (Banerjee et al., 2016). The approach is now being extended at scale in multiple state education systems.

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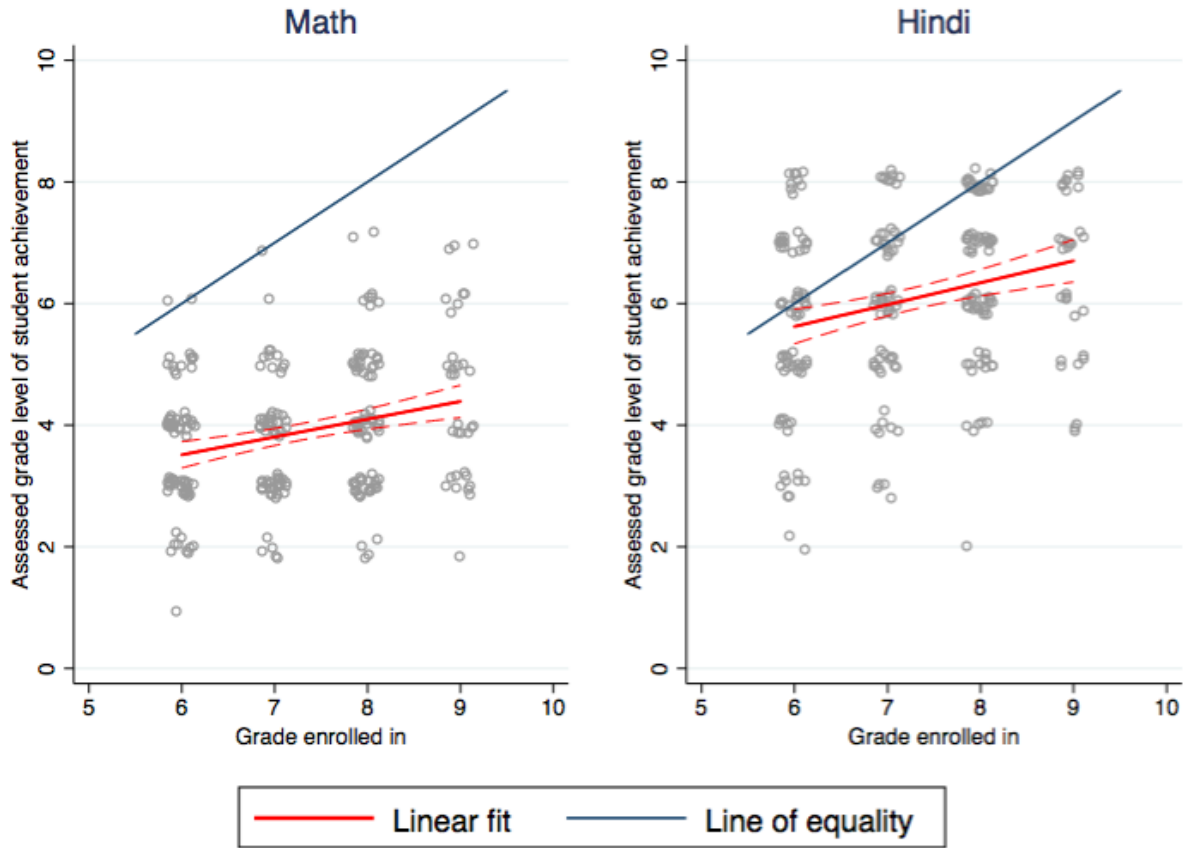
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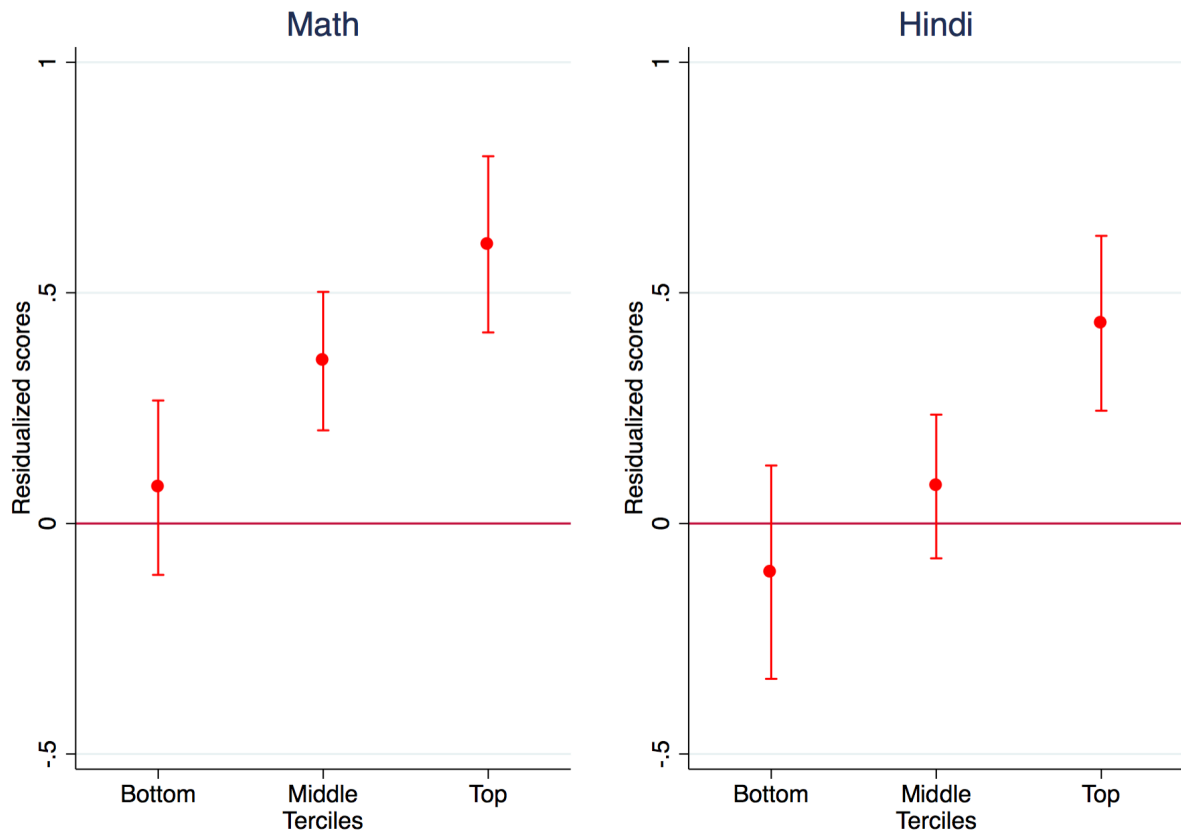
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Figure 1: Assessed levels of student achievement vs. current grade enrolled in school



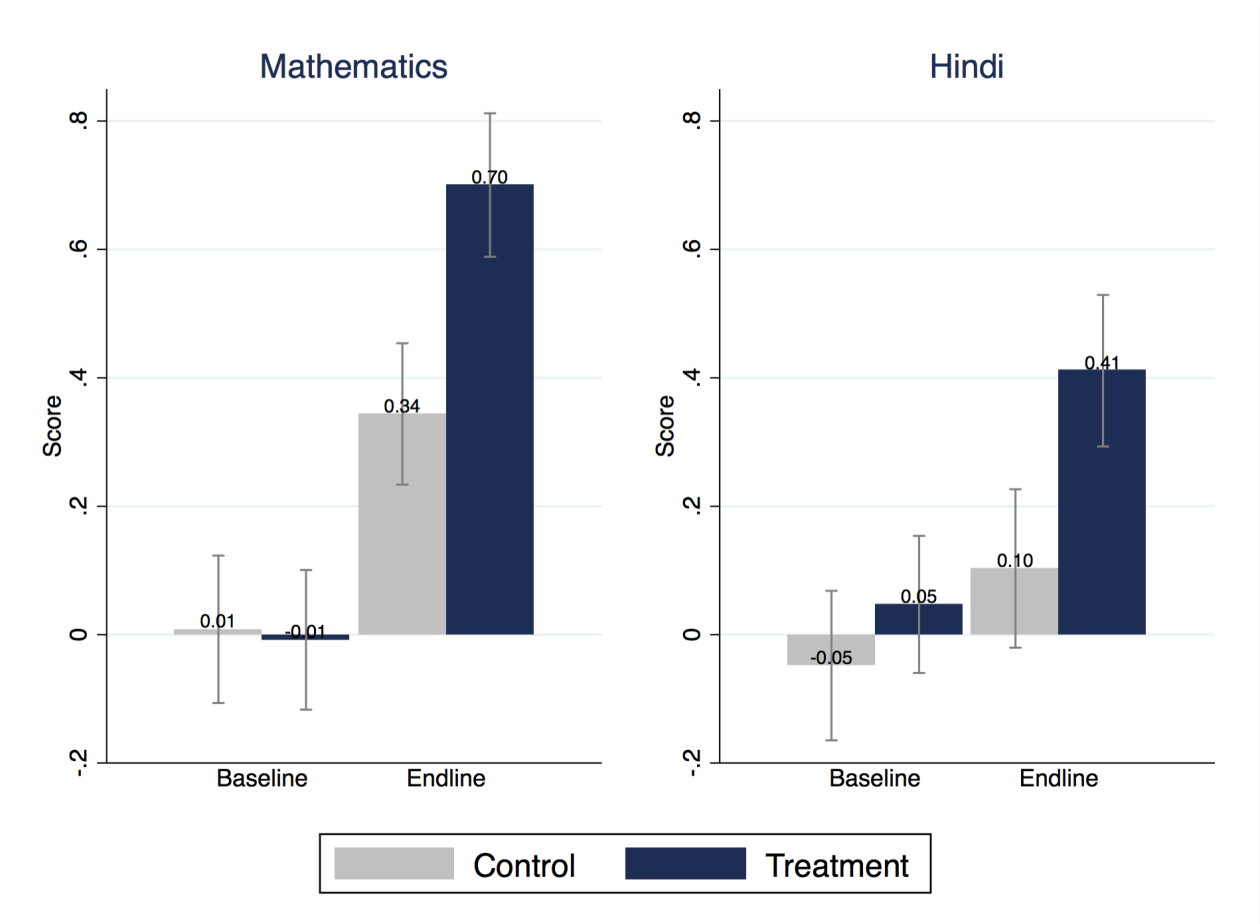
Note: This figure shows, for treatment group, the estimated level of student achievement (determined by the Mindspark CAL program) plotted against the grade they are enrolled in. These data are from the *initial* diagnostic test, and do not reflect any instruction provided by Mindspark. In both subjects, we find three main patterns: (a) there is a general deficit between average attainment and grade-expected norms; (b) this deficit is larger in later grades and (c) within each grade, there is a wide dispersion of student achievement.

Figure 2: Business-as-usual progress in learning



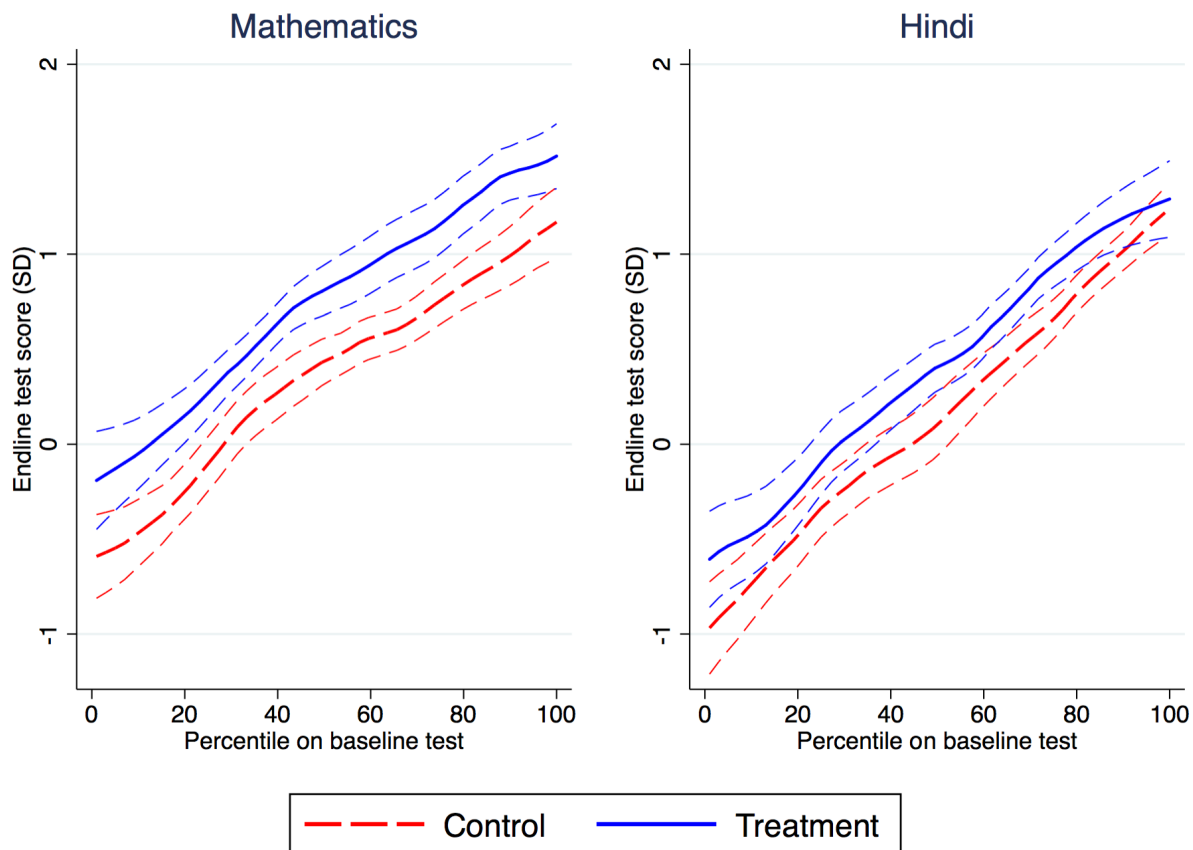
Note: This figure shows the value-added in the control group for students in different terciles of the within-grade achievement distribution. Value-added is measured on our independently-administered tests at baseline and endline tests in September 2015 and February 2016 respectively.

Figure 3: Mean difference in test scores between lottery winners and losers



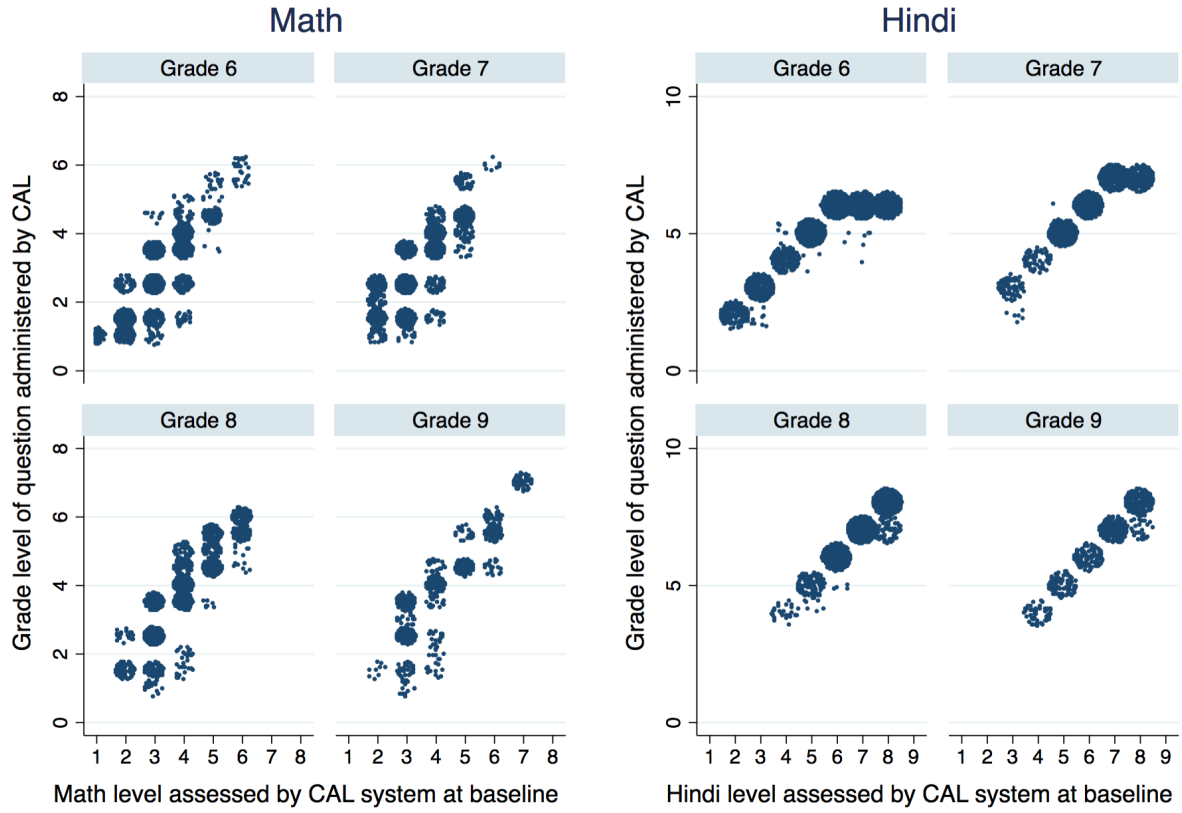
Note: This figure shows mean of test scores, normalized with reference to baseline, across treatment and control groups in the two rounds of testing with 95% confidence intervals. Test scores were linked within-subject through IRT models, pooling across grades and across baseline and endline, and are normalized to have a mean of zero and a standard deviation of one in the baseline. Whereas baseline test scores were balanced between lottery-winners and lottery-losers, endline scores are significantly higher for the treatment group.

Figure 4: Non-parametric investigation of treatment effects by baseline percentiles



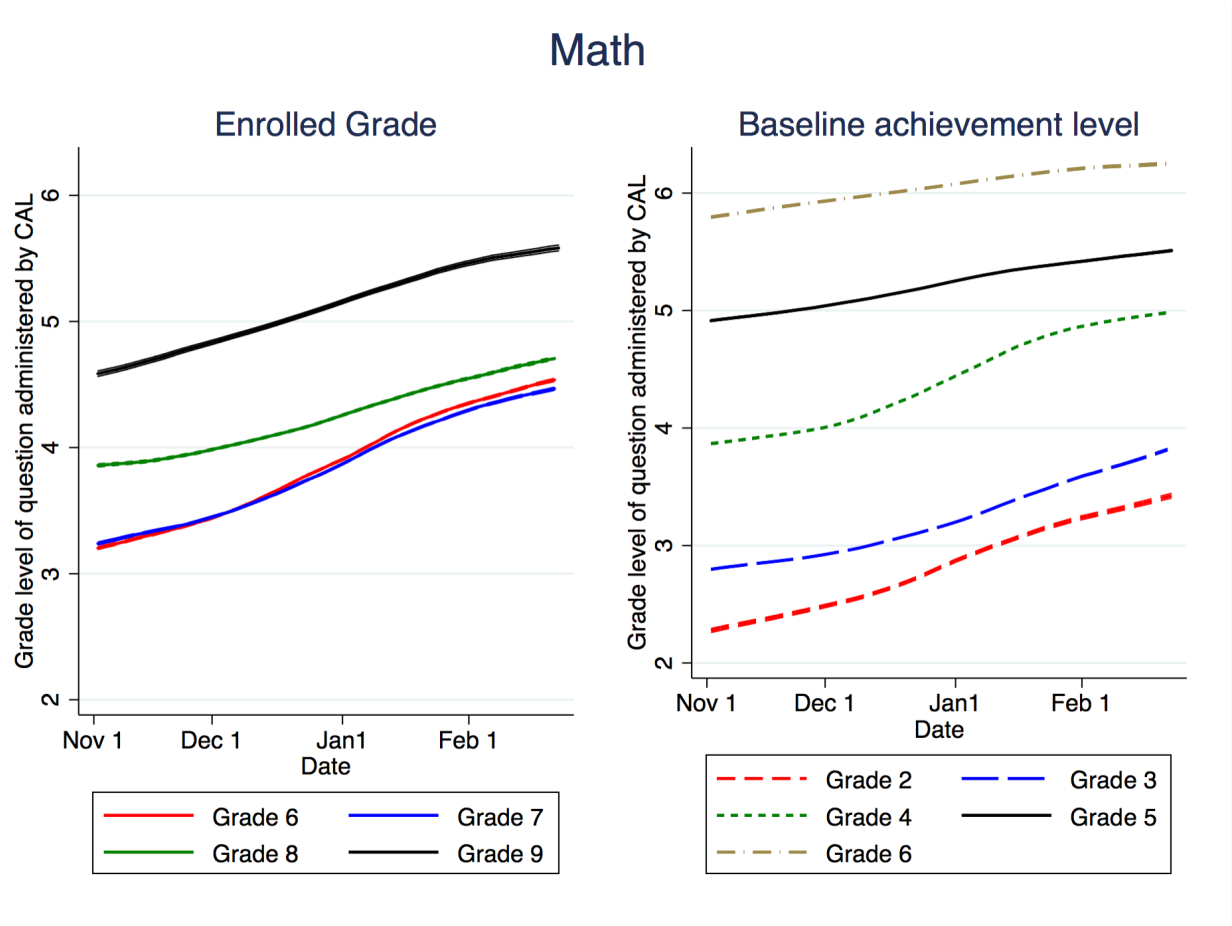
Note: The figures present kernel-weighted local mean smoothed plots which relate endline test scores to percentiles in the baseline achievement, separately for the treatment and control groups, alongside 95% confidence intervals. At all percentiles of baseline achievement, treatment group students see larger gains over the study period than the control group, with no strong evidence of differential absolute magnitudes of gains across the distribution.

Figure 5: Precise customization of instruction by the Mindspark CAL program



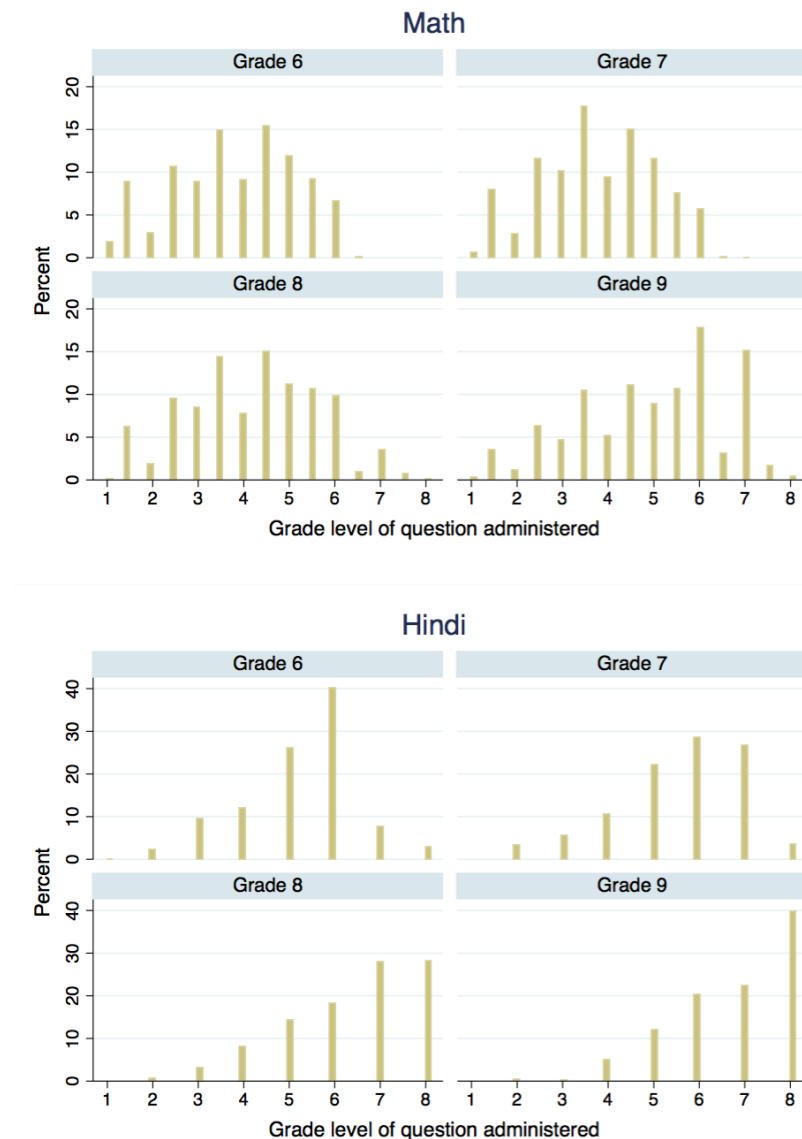
Note: This figure shows, for treatment group, the grade level of questions administered by the computer adaptive system to students on a single day near the beginning of the intervention. In each grade of enrolment, actual level of student attainment estimated by the CAL software differs widely; this wide range is covered through the customization of instructional content by the CAL software.

Figure 6: Dynamic updating and individualization of content in Mindspark



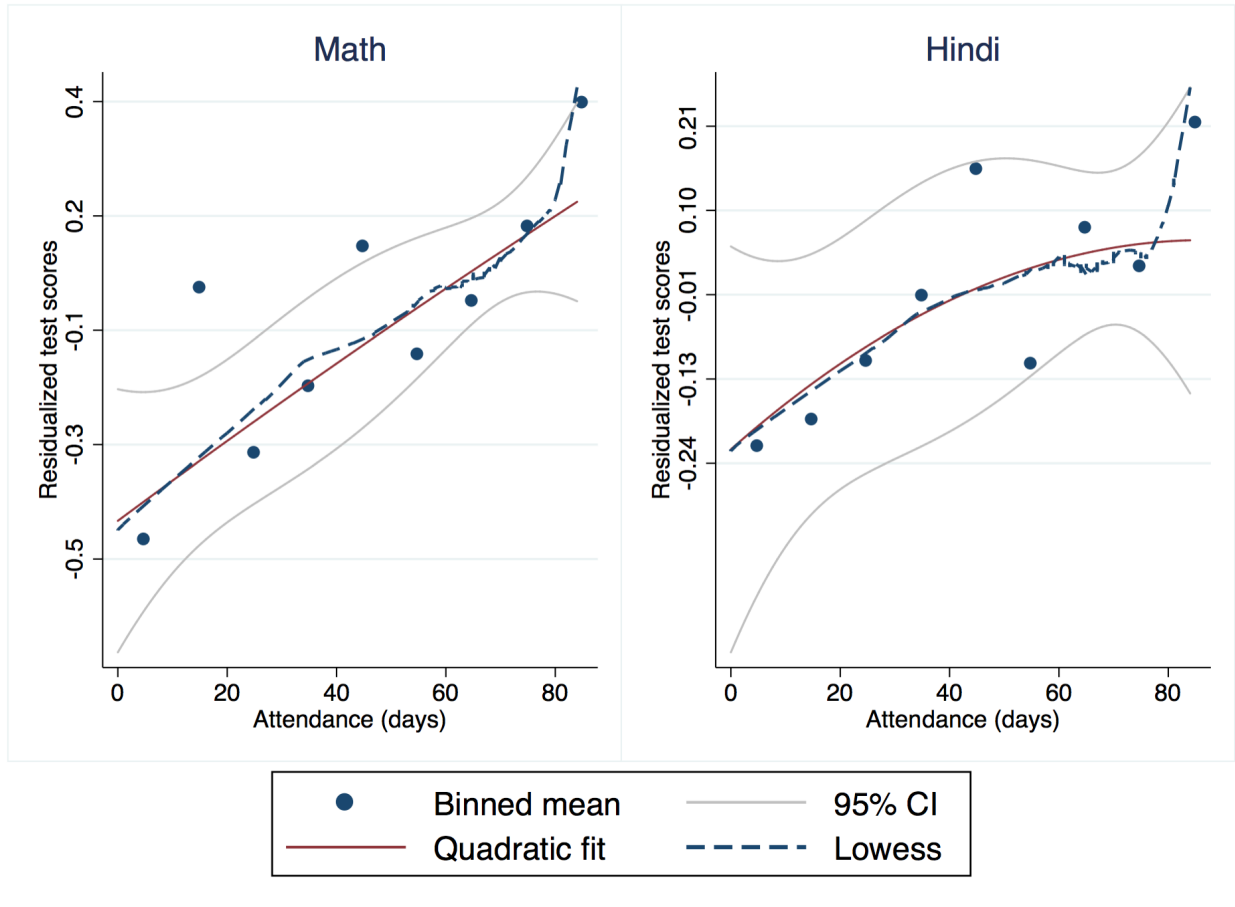
Note: This figure shows kernel-weighted local mean smoothed lines relating the level of difficulty of the math questions administered to students in the treatment group with the date of administration. The left panel presents separate lines by the actual grade of enrolment. The right panel presents separate lines by the level of achievement assessed at baseline by the CAL software. Please note 95% confidence intervals are plotted as well but, given the large data at our disposal, estimates are very precise and the confidence intervals are narrow enough to not be visually discernible.

Figure 7: Distribution of questions administered by Mindspark CAL system



Note: The two panels above show the distribution, by grade-level, of the questions that were administered by the Mindspark CAL system over the duration of treatment in both math and Hindi. Note that in math, students received very few questions at the level of the grade they are enrolled in; this reflects the system's diagnosis of their actual learning levels. In Hindi, by contrast, students received a significant portion of instruction at grade-level competence which is consistent with the initial deficits in achievement in Hindi being substantially smaller than in math (see Fig. 1).

Figure 8: Dose response relationship



Note: This figure explores the relationship between value-added and attendance in the Mindspark program among the lottery-winners. It presents the mean value-added in bins of attendance along with a quadratic fit and a lowess smoothed non-parametric plot.

Table 1: Sample descriptives and balance on observables

	Mean (treatment)	Mean (control)	Difference	SE	N (treatment)	N (control)
<u>Panel A: All students in the baseline sample</u>						
<i>Demographic characteristics</i>						
Female	0.76	0.76	0.00	0.03	314	305
Age (years)	12.68	12.48	0.20	0.13	306	296
SES index	0.00	0.05	-0.05	0.14	314	305
<i>Grade in school</i>						
Grade 4	0.01	0.01	-0.00	0.01	305	299
Grade 5	0.01	0.02	-0.01	0.01	305	299
Grade 6	0.27	0.30	-0.04	0.04	305	299
Grade 7	0.26	0.26	0.00	0.04	305	299
Grade 8	0.30	0.28	0.02	0.04	305	299
Grade 9	0.15	0.13	0.02	0.03	305	299
<i>Baseline test scores</i>						
Math	-0.01	0.01	-0.02	0.08	313	304
Hindi	0.05	-0.05	0.10	0.08	312	305
Present at endline	0.838	0.885	-0.048*	0.028	314	305
<u>Panel B: Only students present in Endline</u>						
<i>Demographic characteristics</i>						
Female	0.77	0.76	0.01	0.04	263	270
Age (years)	12.60	12.46	0.13	0.14	257	263
SES index	-0.10	0.04	-0.14	0.14	263	270
<i>Grade in school</i>						
Grade 4	0.01	0.01	-0.00	0.01	255	266
Grade 5	0.01	0.02	-0.01	0.01	255	266
Grade 6	0.29	0.31	-0.02	0.04	255	266
Grade 7	0.25	0.25	0.00	0.04	255	266
Grade 8	0.30	0.29	0.02	0.04	255	266
Grade 9	0.14	0.12	0.02	0.03	255	266
<i>Baseline test scores</i>						
Math	-0.03	-0.02	-0.02	0.09	262	269
Hindi	0.06	-0.07	0.13	0.08	263	270

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Treatment and control here refer to groups who were randomly assigned to receive an offer of Mindspark voucher till March 2016. Variables used in this table are from the baseline data collection in September 2015. The data collection consisted of two parts: (a) a self-administered student survey, from which demographic characteristics, details of schooling and private tutoring are taken and (b) assessment of skills in math and Hindi, administered using pen-and-paper tests. Tests were designed to cover wide ranges of achievement and to be linked across grades, as well as between baseline and endline assessments, using common items. Scores are scaled here using Item Response theory models and standardized to have a mean of zero and standard deviation of one in the baseline. The SES index refers to a wealth index generated using the first factor from a Principal Components Analysis consisting of indicators for ownership of various consumer durables and services in the household.

Table 2: Intent-to-treat (ITT) Effects in a regression framework

	(1)	(2)	(3)	(4)
	Dep var: Standardized IRT scores (endline)			
	Math	Hindi	Math	Hindi
Treatment	0.36*** (0.063)	0.22*** (0.076)	0.36*** (0.062)	0.22*** (0.064)
Baseline score	0.54*** (0.047)	0.67*** (0.034)	0.55*** (0.039)	0.69*** (0.039)
Constant	0.36*** (0.031)	0.15*** (0.038)	0.36*** (0.043)	0.15*** (0.045)
Strata fixed effects	Y	Y	N	N
Observations	529	533	529	533
R-squared	0.392	0.451	0.392	0.465

Note: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$ Treatment is a dummy variable indicating a randomly-assigned offer of Mindspark voucher till March 2016. Tests in both math and Hindi were designed to cover wide ranges of achievement and to be linked across grades, as well as between baseline and endline assessments, using common items. Scores are scaled here using Item Response theory models and standardized to have a mean of zero and standard deviation of one in the baseline.

Table 3: Treatment effect by specific competence assessed

(a) Mathematics

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Dep var: Proportion of questions answered correctly</i>							
	Arithmetic computation	Word problems - computation	Data interpretation	Fractions and decimals	Geometry and Measurement	Numbers	Pattern recognition
Treatment	0.078*** (0.016)	0.071*** (0.016)	0.044** (0.020)	0.072*** (0.020)	0.14*** (0.026)	0.15*** (0.023)	0.11*** (0.029)
Baseline math score	0.13*** (0.0070)	0.11*** (0.0095)	0.080*** (0.013)	0.090*** (0.011)	0.050*** (0.014)	0.067*** (0.012)	0.094*** (0.013)
Constant	0.66*** (0.0080)	0.50*** (0.0077)	0.38*** (0.0098)	0.33*** (0.010)	0.39*** (0.013)	0.45*** (0.011)	0.36*** (0.015)
Observations	531	531	531	531	531	531	531
R-squared	0.365	0.227	0.095	0.153	0.092	0.134	0.109

(b) Hindi

	(1)	(2)	(3)	(4)
<i>Dep var: Proportion of questions answered correctly</i>				
	Sentence completion	Retrieve explicitly stated information	Make straightforward inferences	Interpret and integrate ideas and information
Treatment	0.047* (0.024)	0.046*** (0.016)	0.064*** (0.022)	0.055*** (0.016)
Baseline Hindi score	0.13*** (0.016)	0.14*** (0.0079)	0.14*** (0.011)	0.064*** (0.013)
Constant	0.73*** (0.012)	0.59*** (0.0078)	0.52*** (0.011)	0.31*** (0.0079)
Observations	533	533	533	533
R-squared	0.186	0.382	0.305	0.132

Note: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The tables above show the impact of the treatment on specific competences. The dependent variable in each regression is the proportion of questions related to the competence that a student answered correctly. Baseline scores are IRT scores in the relevant subject from the baseline assessment. Treatment is a dummy variable indicating a randomly-assigned offer of Mindspark voucher till March 2016. All regressions include randomization strata fixed effects.

Table 4: Heterogeneity in treatment effect by gender, socio-economic status and initial test scores

	(1)	(2)	(3)	(4)	(5)	(6)
<i>Dep var: Standardized IRT scores (endline)</i>						
VARIABLES	Math	Hindi	Math	Hindi	Math	Hindi
Treatment	0.43*** (0.14)	0.22** (0.10)	0.36*** (0.063)	0.24*** (0.067)	0.36*** (0.064)	0.22*** (0.076)
Female	-0.032 (0.15)	0.17 (0.16)				
SES index			0.0095 (0.029)	0.088*** (0.020)		
Baseline score	0.54*** (0.047)	0.67*** (0.034)	0.54*** (0.045)	0.64*** (0.032)	0.51*** (0.057)	0.67*** (0.044)
Treatment * Female	-0.082 (0.14)	-0.0037 (0.13)				
Treatment * SES index			-0.0011 (0.044)	0.016 (0.042)		
Treatment * Baseline score					0.058 (0.075)	-0.0025 (0.078)
Constant	0.38*** (0.11)	0.021 (0.11)	0.36*** (0.031)	0.15*** (0.033)	0.36*** (0.031)	0.15*** (0.037)
Observations	529	533	529	533	529	533
R-squared	0.393	0.453	0.393	0.472	0.393	0.451

Note: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$ Treatment is a dummy variable indicating a randomly-assigned offer of Mindspark voucher till March 2016. The SES index and test scores are defined as in Tables 1 and 2 respectively. All regressions include strata fixed effects.

Table 5: Heterogeneity in treatment effect by within-grade terciles

	(1)	(2)
	<i>Dep var: Standardized IRT scores (endline)</i>	
VARIABLES	Math	Hindi
Bottom Tercile	0.14 (0.091)	-0.11 (0.10)
Middle Tercile	0.35*** (0.073)	0.11 (0.078)
Top Tercile	0.57*** (0.086)	0.46*** (0.079)
Treatment	0.36*** (0.11)	0.34*** (0.13)
Treatment*Middle Tercile	0.081 (0.15)	-0.21 (0.17)
Treatment*Top Tercile	-0.040 (0.16)	-0.16 (0.15)
Baseline test score	0.41*** (0.058)	0.53*** (0.061)
Observations	529	533
R-squared	0.555	0.516

Note: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Treatment is a dummy variable indicating a randomly-assigned offer of Mindspark voucher till March 2016. Test scores are scaled as in Table 2.

Table 6: Treatment effect on items linked to grade levels

	(1)	(2)	(3)	(4)
	<i>Dep var: Proportion of questions answered correctly</i>			
	<u>Math</u>		<u>Hindi</u>	
VARIABLES	At or above grade level	Below grade level	At or above grade level	Below grade level
Treatment	0.0023 (0.039)	0.082*** (0.012)	0.069** (0.024)	0.051*** (0.013)
Baseline math score	0.044 (0.025)	0.095*** (0.0056)		
Baseline Hindi score			0.11*** (0.016)	0.13*** (0.0065)
Constant	0.31*** (0.018)	0.49*** (0.0058)	0.44*** (0.012)	0.58*** (0.0065)
Observations	286	505	287	507
R-squared	0.025	0.341	0.206	0.379

Note: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The tables above show the impact of the treatment on questions below or at/above grade levels for individual students. The dependent variable in each regression is the proportion of questions that a student answered correctly. Our endline assessments were designed to be informative at students' actual levels of achievement and therefore did not include many items at grade 8 level and above. Therefore students in Grades 8 and 9 are not included in regressions on items at/above grade level. Baseline scores are IRT scores in the relevant subject from the baseline assessment. Treatment is a dummy variable indicating a randomly-assigned offer of Mindspark voucher till March 2016. All regressions include randomization strata fixed effects.

Table 7: Treatment effect on school exams

	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES	Hindi	<u>Dep var: Standardized test scores</u>				
		Math	Science	Social Sciences	English	Aggregate
Treatment	0.19** (0.089)	0.058 (0.076)	0.077 (0.092)	0.10 (0.11)	0.080 (0.10)	0.097 (0.080)
Baseline Hindi score	0.48*** (0.094)		0.28*** (0.064)	0.41*** (0.098)	0.29*** (0.069)	0.33*** (0.061)
Baseline math score		0.29*** (0.039)	0.10** (0.036)	0.25*** (0.052)	0.11** (0.049)	0.16*** (0.037)
Constant	0.40 (1.01)	0.14 (0.50)	0.88** (0.39)	0.69 (0.69)	1.11 (0.66)	0.68 (0.56)
Observations	595	594	593	592	595	595
R-squared	0.188	0.069	0.117	0.173	0.137	0.202

Note: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. This table shows the effect of receiving the Mindspark voucher on the final school exams, held in March 2016 after the completion of the intervention. Treatment is a dummy variable indicating a randomly-assigned offer of Mindspark voucher till March 2016. Test scores in the school exams are normalized within school*grade to have a mean of zero and a standard deviation of one in the control group. All regressions include grade and school fixed effects. For math and Hindi, we have included baseline scores in the same subject from our independent baseline tests; for other subjects, where we did not administer subject-specific baseline tests, we include test scores from both subjects.

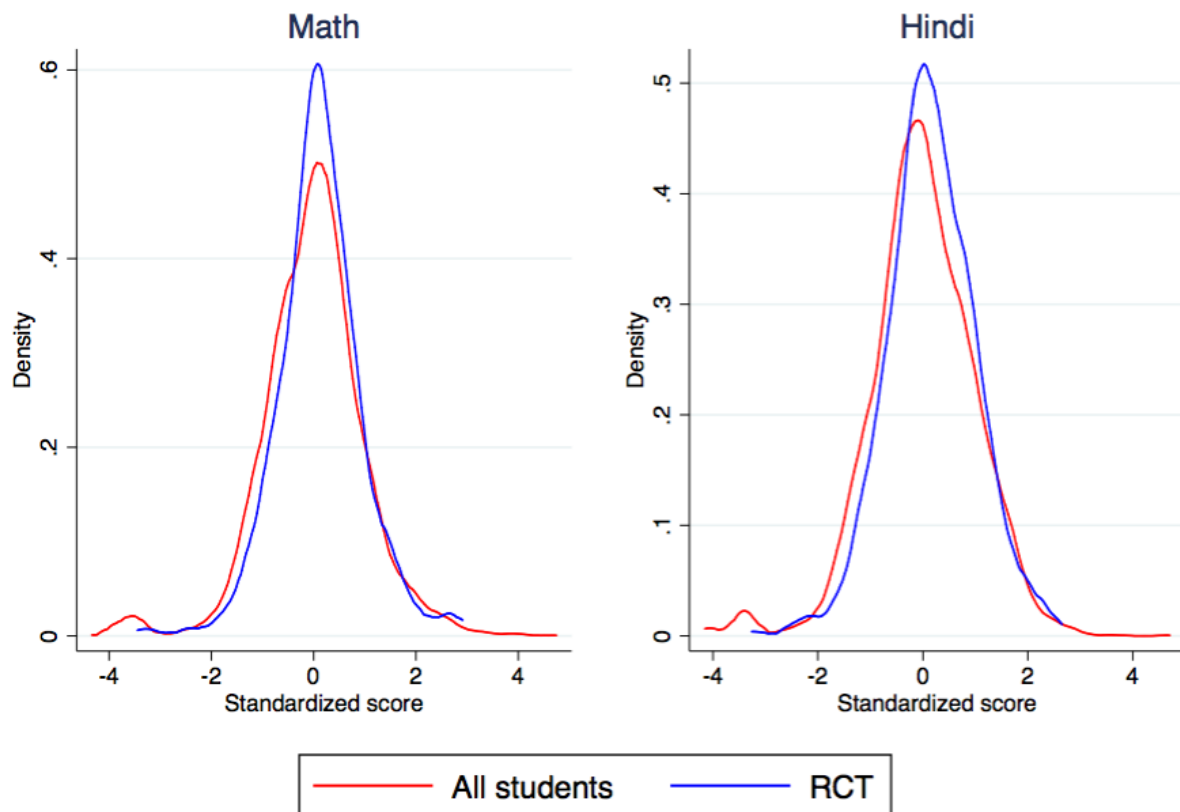
Table 8: Dose-response of Mindspark attendance

	(1)	(2)	(3)	(4)	(5)	(6)
	<i>Dep var: Standardized IRT scores (endline)</i>					
VARIABLES	IV estimates		OLS VA (full sample)		OLS VA (Treatment group)	
	Math	Hindi	Math	Hindi	Math	Hindi
Attendance (days)	0.0065*** (0.0011)	0.0040*** (0.0011)	0.0068*** (0.00087)	0.0037*** (0.00090)	0.0075*** (0.0018)	0.0033* (0.0020)
Baseline score	0.53*** (0.036)	0.67*** (0.037)	0.54*** (0.039)	0.69*** (0.039)	0.57*** (0.062)	0.68*** (0.056)
Constant			0.35*** (0.040)	0.16*** (0.042)	0.31*** (0.12)	0.18 (0.13)
Observations	529	533	529	533	261	263
R-squared	0.422	0.460	0.413	0.468	0.413	0.429
Angrist-Pischke F-statistic for weak instrument	1238	1256				
Diff-in-Sargan statistic for exogeneity (p-value)	0.26	0.65				
Extrapolated estimates of 90 days' treatment (SD)	0.585	0.36	0.612	0.333	0.675	0.297

Note: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$ Treatment group students who were randomly-selected for the Mindspark voucher offer but who did not take up the offer have been marked as having 0% attendance, as have all students in the control group. Columns (1) and (2) instrument attendance in Mindspark with the randomized allocation of a scholarship and include randomization strata fixed effects, Columns (3) and (4) present OLS value-added models in the full sample, Columns (5) and (6) present OLS value-added models using only data on the lottery-winners. Scores are scaled here using Item Response theory models and linked across grades and across baseline and endline assessments using common anchor items. Tests in both math and Hindi are standardized to have a mean of zero and standard deviation of one in the baseline.

Appendix A Additional figures and tables

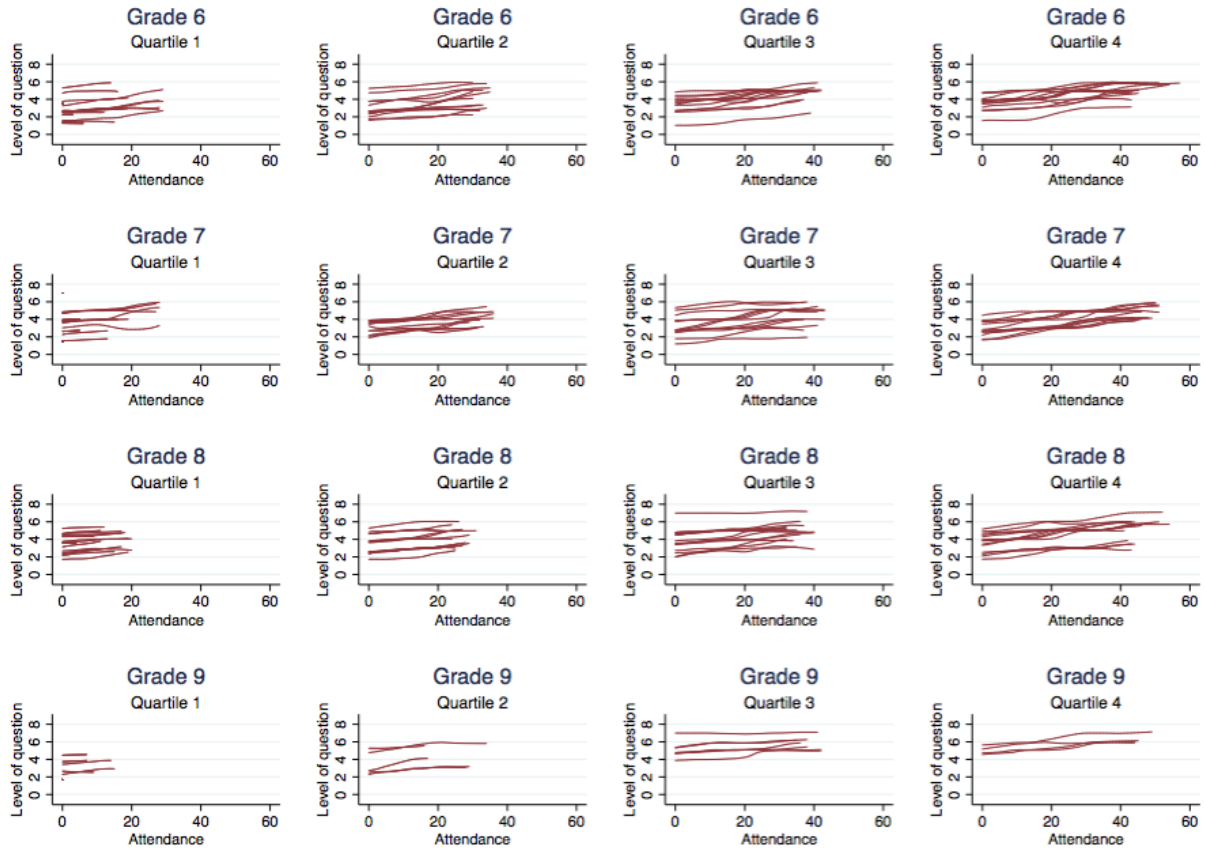
Figure A.1: Comparing pre-program achievement of study participants and non-participants



403 study children matched to school records of 2014-15

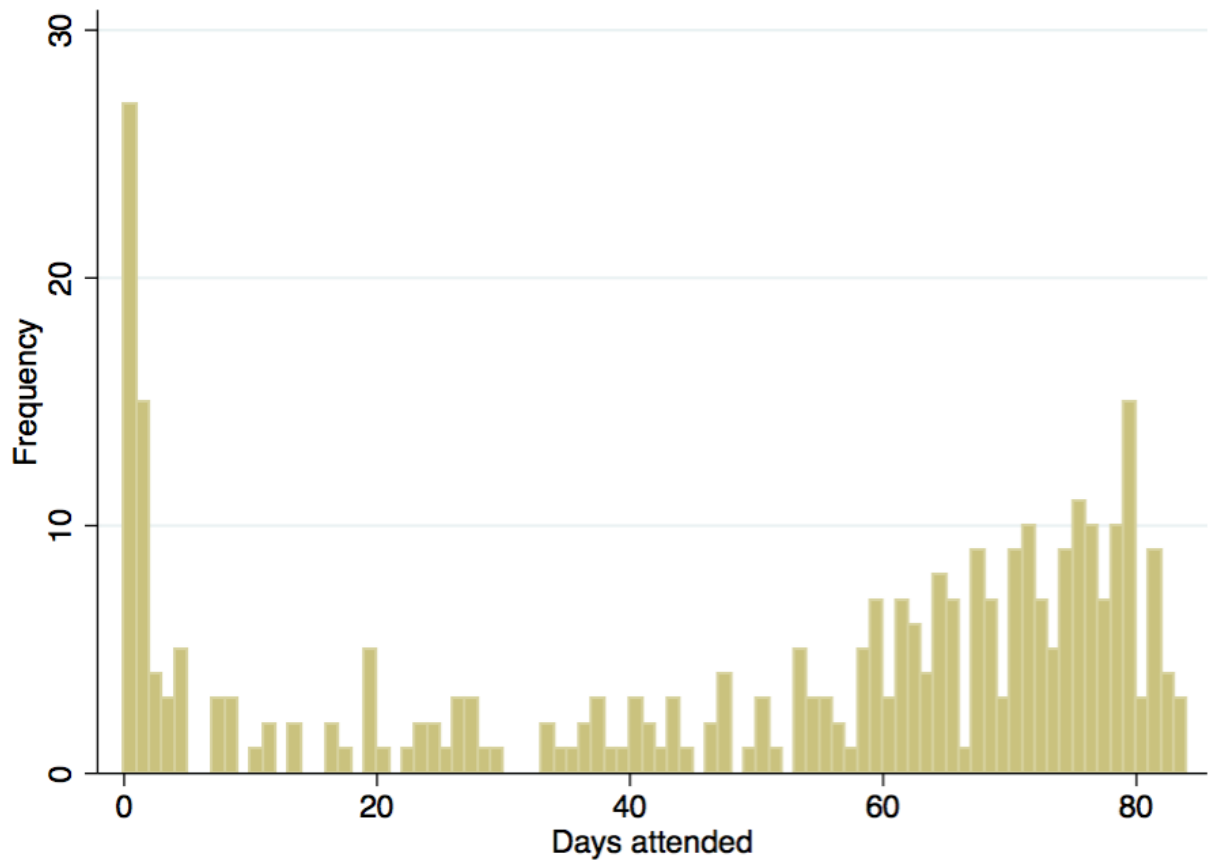
Note: The panels compare the final scores for the 2014-15 school year, i.e. the pre-program academic year, for study participants and non-participants. Test scores have been standardized within school*grade cells. The study participants are positively selected into the RCT in comparison to their peers but the magnitude of selection is modest and there is near-complete common support between the two groups in pre-program academic achievement. See Table A.1 for further details.

Figure A.2: Learning trajectories of individual students in the treatment group



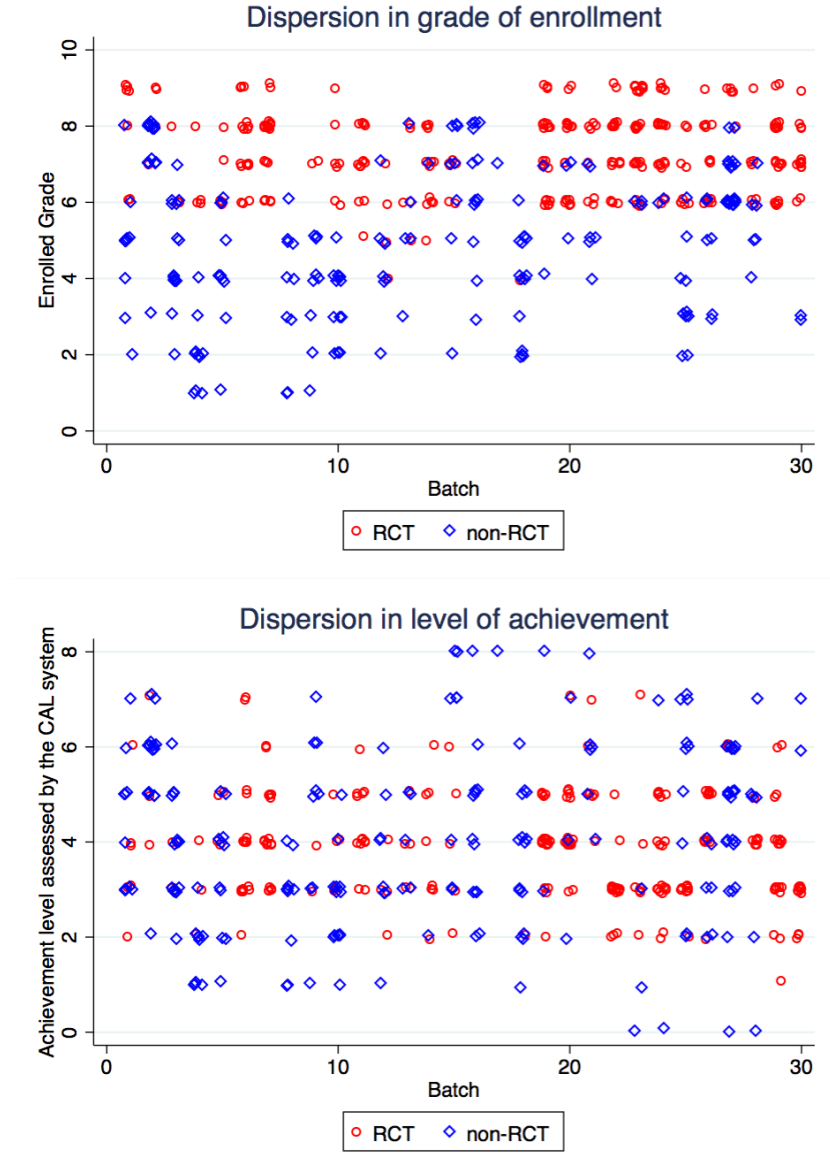
Note: Each line in the panels above is a local mean smoothed plot of the grade level of questions administered in Mathematics by the computer adaptive system against the days that the student utilized the Mindspark math software (Attendance). The panels are organized by the grade of enrolment and the within-grade quartile of attendance in Mindspark.

Figure A.3: Distribution of take-up among lottery-winners



Note: This figure shows the distribution of attendance in the Mindspark centers among the lottery-winners. Over the study period, the Mindspark centers were open for 86 working days.

Figure A.4: Composition of group instruction batches in Mindspark centers



Note: The two panels above show the composition of batches in Mindspark centers by the grade students are enrolled in and by their level of math achievement, as assessed by the Mindspark CAL system. We separately identify students in the treatment group from fee-paying students who were not part of the study but were part of the small group instruction in each batch. Note that, while our study is focused on students from grades 6-9, the centers cater to students from grades 1-8. Batches are chosen by students based on logistical convenience and hence there is substantial variation in grade levels and student achievement within each batch with little possibility of achievement-based tracking. This confirms that it would not have been possible to customize instruction in the instructor-led small group instruction component of the intervention.

Table A.1: Comparing pre-program exam results of study participants and non-participants

	RCT	Non-study	Difference	SE	N(RCT)	N(non-study)
Math	0.13	-0.01	0.14***	0.05	409	4067
Hindi	0.16	-0.02	0.17***	0.05	409	4067
Science	0.09	-0.01	0.10**	0.05	409	4067
Social Science	0.13	-0.01	0.15***	0.05	409	4067
English	0.14	-0.01	0.15***	0.05	409	4067

Note: This table presents the mean scores of study participants and non-participants, standardized within each school*grade, in the 2014-15 school year. Study participants are, on average, positively selected compared to their peers.

Table A.2: ITT estimates with within-grade normalized test scores

VARIABLES	(1)	(2)	(3)	(4)
	Dep var: Endline scores			
	Math	Hindi	Math	Hindi
Treatment	0.37*** (0.067)	0.21*** (0.067)	0.36*** (0.068)	0.21*** (0.073)
Baseline math score	0.56*** (0.042)		0.55*** (0.050)	
Baseline Hindi score		0.70*** (0.040)		0.69*** (0.033)
Constant	0.37*** (0.046)	0.18*** (0.046)	0.37*** (0.033)	0.18*** (0.036)
Observations	517	521	517	521
R-squared	0.375	0.459	0.376	0.457
Strata fixed effects			Y	Y

Note: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$ Treatment is a dummy variable indicating a randomly-assigned offer of Mindspark voucher till March 2016. The SES index refers to a wealth index generated using the first factor from a Principal Components Analysis consisting of indicators for ownership of various consumer durables and services in the household. Tests in both math and Hindi were designed to cover wide ranges of ability and to be linked across grades, as well as between baseline and endline assessments, using common items. Scores are scaled here using Item Response theory models and standardized to have a mean of zero and standard deviation of one in the baseline in each grade.

Table A.3: Correlates of attendance

VARIABLES	(1)	(2)	(3)	(4)
	Attendance (days)			
Female	3.81 (3.90)	2.51 (3.93)	2.89 (3.89)	4.00 (3.90)
SES index	-3.26*** (1.04)	-3.49*** (1.07)	-3.43*** (1.06)	-3.19*** (1.06)
Attends math tuition			-1.95 (4.41)	0.62 (4.53)
Attends Hindi tuition			7.27* (4.38)	5.32 (4.50)
Baseline math score		-1.07 (2.05)	-0.99 (2.11)	-0.59 (2.09)
Baseline Hindi score		3.66* (2.06)	4.17** (2.10)	5.49*** (2.10)
Constant	46.8*** (3.39)	47.7*** (3.42)	45.5*** (3.79)	43.9*** (3.79)
Grade Fixed Effects	N	N	N	Y
Observations	313	310	310	301
R-squared	0.036	0.045	0.057	0.120

Note: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

This table shows correlates of days attended in the treatment group i.e. lottery-winners who had been offered a Mindspark voucher. Students from poorer backgrounds, and students with higher baseline achievement in Hindi, appear to have greater attendance but the implied magnitudes of these correlations are small. A standard deviation increase in the SES index is associated with a decline in attendance by about 3 days, and a standard deviation increase in Hindi baseline test scores is associated with an additional 5 days of attendance. We find no evidence of differential attendance by gender or by baseline math score.

Table A.4: Quadratic dose-response relationship

	(1)	(2)	(3)	(4)
	Full sample		Treatment group	
	Math	Hindi	Math	Hindi
Attendance (days)	0.0056 (0.0054)	0.0064 (0.0058)	0.0079 (0.0073)	0.0064 (0.0083)
Attendance squared	0.000016 (0.000073)	-0.000037 (0.000078)	-5.52e-06 (0.000084)	-0.000037 (0.000094)
Baseline math score	0.54*** (0.039)		0.57*** (0.062)	
Baseline Hindi score		0.69*** (0.039)		0.68*** (0.057)
Constant	0.35*** (0.041)	0.15*** (0.043)	0.30** (0.14)	0.15 (0.16)
Observations	529	533	261	263
R-squared	0.413	0.468	0.413	0.429

Note: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. This table models the dose-response relationship between Mindspark attendance and value-added quadratically. Results are estimated using OLS in the full sample and the treatment group only.

Table A.5: Dose-response of Mindspark attendance

	(1)	(2)	(3)	(4)	(5)	(6)
	<i>Dep var:</i> Standardized IRT scores (endline)					
VARIABLES	OLS VA (full sample) Math	Hindi	IV models (full sample) Math	Hindi	OLS VA (Treatment group) Math	Hindi
Days of Math instruction	0.018*** (0.0023)		0.017*** (0.0028)		0.020*** (0.0047)	
Days of Hindi instruction		0.011*** (0.0026)		0.011*** (0.0032)		0.0096* (0.0055)
Baseline score	0.54*** (0.039)	0.69*** (0.039)	0.53*** (0.036)	0.67*** (0.037)	0.56*** (0.061)	0.68*** (0.056)
Constant	0.35*** (0.040)	0.16*** (0.042)			0.30*** (0.12)	0.18 (0.13)
Observations	529	533	529	533	261	263
R-squared	0.414	0.469	0.423	0.459	0.414	0.430
Angrist-Pischke F-statistic for weak instrument			1243	1100		
Diff-in-Sargan statistic for exogeneity (p-value)			0.21	0.87		
Extrapolated estimates of 45 days' treatment (SD)	0.81	0.495	0.765	0.495	0.90	0.432

Note: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$ Treatment group students who were randomly-selected for the Mindspark voucher offer but who did not take up the offer have been marked as having 0% attendance, as have all students in the control group. Days attended in Math/Hindi are defined as the number of sessions of either CAL or small group instruction attended in that subject, divided by two. Columns (1) and (2) present OLS value-added models for the full sample, Columns (3) and (4) present IV regressions which instrument attendance with the randomized allocation of a voucher and include fixed effects for randomization strata, and Columns (5) and (6) present OLS value-added models using only data on the lottery-winners. Scores are scaled here as in Table 2.

Table A.6: ITT estimates with inverse probability weighting

	(1)	(2)	(3)	(4)
	Dep var: Endline test scores			
VARIABLES	Math	Hindi	Math	Hindi
Treatment	0.37*** (0.062)	0.22*** (0.064)	0.37*** (0.061)	0.23*** (0.063)
Baseline subject score	0.55*** (0.039)	0.68*** (0.040)	0.54*** (0.037)	0.66*** (0.038)
Constant	0.36*** (0.043)	0.16*** (0.045)	0.36*** (0.042)	0.16*** (0.043)
Strata fixed effects			Y	Y
Observations	529	531	529	531
R-squared	0.393	0.455	0.442	0.504

Note: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$ Treatment is a dummy variable indicating a randomly-assigned offer of Mindspark voucher till March 2016. Results in this table are weighted by the inverse of the predicted probability of having scores in both math and Hindi in the endline; the probability is predicted using a probit model with baseline subject scores, sex of the child, SES index and dummies for individual Mindspark centers as predictors. Tests in both math and Hindi were designed to cover wide ranges of ability and to be linked across grades, as well as between baseline and endline assessments, using common items. Scores are scaled here using Item Response theory models and standardized to have a mean of zero and standard deviation of one in the baseline in each grade.

Table A.7: Lee bounds estimates of ITT effects

	(1) Math	(2) Hindi
Lower	0.267 (0.119)	0.21 (0.127)
Upper	0.445 (0.093)	0.41 (0.102)
Lower 95% CI	0.071	0.0004
Upper 95% CI	0.599	0.58

Note: Analytic standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$ This table presents Lee (2009) bounds on the ITT effects of winning a voucher in both math and Hindi. The bounds are tightened using dummy variables for the Mindspark centers. Tests in both math and Hindi were designed to cover wide ranges of ability and to be linked across grades, as well as between baseline and endline assessments, using common items. Scores are scaled here using Item Response theory models and standardized to have a mean of zero and standard deviation of one in the baseline in each grade.

Table A.8: ITT estimates, by source of test item

	(1)	(2)	(3)	(4)
	Dep var: Proportion correct in endpoint			
	Math		Hindi	
VARIABLES	El items	non-El items	El items	non-El items
Treatment	0.10*** (0.013)	0.071*** (0.010)	0.050*** (0.017)	0.042*** (0.011)
Baseline score	0.094*** (0.0096)	0.096*** (0.0073)	0.14*** (0.0086)	0.12*** (0.0058)
Constant	0.46*** (0.0067)	0.47*** (0.0049)	0.61*** (0.0083)	0.48*** (0.0056)
Observations	531	531	533	533
R-squared	0.228	0.346	0.308	0.403

Note: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$ Treatment is a dummy variable indicating a randomly-assigned offer of a Mindspark voucher till March 2016. Tests in both math and Hindi were assembled using items from different international and Indian assessments, some of which were developed by EI. EI developed assessments include the Student Learning Survey, the Quality Education Study and the Andhra Pradesh Randomized Studies in Education. The dependent variables are defined as the proportion correct on items taken from assessments developed by EI and on other non-EI items. Baseline scores are IRT scores normalized to have a mean of zero and a standard deviation of one.

Table A.9: Treatment effect on take-up of other private tutoring

VARIABLES	(1) Math	(2) Hindi	(3) English	(4) Science	(5) Social Science
Post Sept-2015	0.019* (0.011)	0.018* (0.0096)	0.026*** (0.0098)	0.018** (0.0080)	0.014** (0.0071)
Post * Treatment	0.013 (0.016)	-0.010 (0.012)	-0.0039 (0.013)	0.0017 (0.012)	-0.0056 (0.0086)
Constant	0.21*** (0.0053)	0.13*** (0.0040)	0.18*** (0.0044)	0.14*** (0.0041)	0.098*** (0.0029)
Observations	3,735	3,735	3,735	3,735	3,735
R-squared	0.009	0.004	0.010	0.007	0.005
Number of students	415	415	415	415	415

Note: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. This table shows individual fixed-effects estimates of receiving the Mindspark voucher on the take-up in other private tutoring in various subjects. The dependent variable is whether a child was attending extra tutoring in a given month between July 2015 and March 2016 in the particular subject. This was collected using telephonic interviews with the parents of study students. Observations are at the month*child level. Treatment is a dummy variable indicating a randomly-assigned offer of Mindspark voucher till March 2016.

Appendix B Prior research on hardware and software

Tables B.1 and B.2 offer an overview of experimental and quasi-experimental impact evaluations of interventions providing hardware and software to improve children’s learning. The tables only include studies focusing on students in primary and secondary school (not pre-school or higher education) and only report effects in math and language (not on other outcomes assessed in these studies, e.g., familiarity with computers or socio-emotional skills).

B.1 Selecting studies

This does not intend to be a comprehensive review of the literature. Specifically, we have excluded several impact evaluations of programs (mostly, within education) due to major design flaws (e.g., extremely small sample sizes, having no control group, or dropping attritors from the analysis). These flaws are widely documented in meta-analyses of this literature (see, for example, Murphy et al., 2001; Pearson et al., 2005; Waxman et al., 2003).

We implemented additional exclusions for each table. In Table B.1, we excluded DID designs in which identification is questionable and studies evaluating the impact of subsidies for Internet (for example, Goolsbee and Guryan, 2006). In Table B.2, we excluded impact evaluations of software products for subjects other than math and language or designed to address specific learning disabilities (e.g., dyslexia, speech impairment).

B.2 Reporting effects

To report effect sizes, we followed the following procedure: (a) we reported the difference between treatment and control groups adjusted for baseline performance whenever this was available; (b) if this difference was not available, we reported the simple difference between treatment and control groups (without any covariates other than randomization blocks if applicable); and (c) if neither difference was available, we reported the difference between treatment and control groups adjusted for baseline performance and/or any other covariates that the authors included.

In all RCTs, we reported the intent-to-treat (ITT) effect; in all RDDs and IVs, we reported the local average treatment effect (LATE). In all cases, we only reported the magnitude of effect sizes that were statistically significant at the 5% level. These decisions are non-trivial, as the specifications preferred by the authors of some studies (and reported in the abstracts) are only significant at the 10% level or only become significant at the 5% level after the inclusion of multiple covariates. Otherwise, we mentioned that a program had “no effect” on

the respective subject. Again, this decision is non-trivial because some of these studies were under-powered to detect small to moderate effects.

B.3 Categories in each table

In both tables, we documented the study, the impact evaluation method employed by the authors, the sample, the program, the subject for which the software/hardware was designed to target, and its intensity. Additionally, in Table B.1, we documented: (a) whether the hardware provided included pre-installed software; (b) whether the hardware required any participation from the instructor; and (c) whether the hardware was accompanied by training for teachers. In Table B.2, we documented: (a) whether the software was linked to an official curriculum (and if so, how); (b) whether the software was adaptive (i.e., whether it could *dynamically* adjust the difficulty of questions and/or activities based on students' performance); and (c) whether the software provided *differentiated* feedback (i.e., whether students saw different messages depending on the incorrect answer that they selected).

Table B.1: Impact evaluations of hardware

Study	Method	Sample	Program	Subject	Intensity	Software included?	Instructor's role?	Teacher training?	Effect	Cost
Angrist and Lavy (2002)	IV	Grades 4 and 8, 122 Jewish schools in Israel	Tomorrow-98	Math and language (Hebrew)	Target student-computer ratio of 1:10 in each school	Yes, included educational software from a private company	Not specified	Yes, training for teachers to integrate computers into teaching	Grade 4: -0.4 to -0.3 σ in math and no effect in language	USD 3,000 per machine, including hardware, software, and setup; at 40 computers per school, USD 120,000 per school
Barrera-Osorio and Linden (2009)	RCT	Grades 3-9, 97 public schools in six school districts, Colombia	Computers for Education	Math and language (Spanish)	15 computers per school	Not specified	Use the computers to support children on basic skills (esp. Spanish)	Yes, 20-month training for teachers, provided by a local university	No effect in language or math	Not specified
Malamud and Pop-Eleches (2011)	RDD	Grades 1-12, in six regions, Romania	Euro 200 Program	Math and language (English and Romanian)	One voucher (worth USD 300) towards the purchase of a computer for use at home	Pre-installed software, but educational software provided separately and not always installed	Not specified	Yes, 530 multimedia lessons on the use of computers for educational purposes for students	-0.44 σ in math GPA, -0.56 σ in Romanian GPA, and -0.63 σ in English	Cost of the voucher plus management costs not specified

Cristia et al. (2012)	RCT	319 schools in eight rural areas, Peru	One Laptop per Child	Math and language (Spanish)	One laptop per student and teacher for use at school and home	Yes, 39 applications including: standard applications, educational games, music editing, programming environments, sound and video recording, encyclopedia; also 200 age-appropriate e-books	Not specified	Yes, 40-hour training aimed at facilitating the use of laptops for pedagogical purposes	No effect in math or language	USD 200 per laptop
Mo et al. (2013)	RCT	Grade 3, 13 migrant schools in Beijing, China	One Laptop per Child	Math and language (Chinese)	One laptop per student for use at home	Yes, three sets of software: a commercial, game-based math learning program; a similar program for Chinese; a third program developed by the research team	Not specified	No, but one training session with children and their parents	No effect in math or language	Not specified
Beuermann et al. (2015)	RCT	Grade 2, 28 public schools in Lima, Peru	One Laptop per Child	Math and language (Spanish)	Four laptops (one per student) in each class/section for use at school	Yes, 32 applications including: standard applications, educational games, music editing, programming environments, sound and video recording, encyclopedia	Not specified	No, but weekly training sessions during seven weeks for students	No effect in math or language	USD 188 per laptop

Leuven et al. (2007)	RDD	Grade 8, 150 schools in the Netherlands	Not specified	Math and language (Dutch)	Not specified	Not specified	Not specified	-0.08 SDs in language and no effect in math	This study estimates the effect of USD 90 per pupil for hardware and software
Machin et al. (2007)	IV	Grade 6, 627 (1999-2001) and 810 (2001-2002) primary and 616 (1999-2000) and 714 (2001-2002) secondary schools in England	Not specified	Math and language (English)	Target student-computer ratio of 1:8 in each primary school and 1:5 in each secondary school	Some schools spent funds for ICT for software	Not specified	2.2 pp. increase in the percentage of children reaching minimally acceptable standards in end-of-year exams	This study estimates the effect of doubling funding for ICT (hardware and software) for a Local Education Authority
Fairlie and Robinson (2013)	RCT	Grades 6-10, 15 middle and high public schools in five school districts in California, United States	Not specified	Math and language (English)	One computer per child for use at home	Yes, Microsoft Windows and Office	No	No effect in language or math	Not specified

Table B.2: Impact evaluations of software

Study	Method	Sample	Program	Subject	Intensity	Linked to curriculum?	Dynamically adaptive?	Differentiated feedback?	Effect	Cost
Banerjee et al. (2007)	RCT	Grade 4, 100 municipal schools in Gujarat, India	Year 1: off-the-shelf program developed by Pratham; Year 2: program developed by Media-Pro	Math	120 min./week during or before/after school; 2 children per computer	Gujarati curriculum, focus on basic skills	Yes, question difficulty responds to ability	Not specified	Year 1: 0.35σ on math and no effect in language; Year 2: 0.48σ on math and no effect in language	INR 722 (USD 15.18) per student per year
Linden (2008)	RCT	Grades 2-3, 60 Gyan Shala schools in Gujarat, India	Gyan Shala Computer Assisted Learning (CAL) program	Math	Version 1: 60 min./day during school; Version 2: 60 min./day after school; Both: 2 children per computer (split screen)	Gujarati curriculum, reinforces material taught that day	Not specified	Not specified	Version 1: no effect in math or language; Version 2: no effect in math or language	USD 5 per student per year
Carrillo et al. (2010)	RCT	Grades 3-5, 16 public schools in Guayaquil, Ecuador	Personalized Complementary and Interconnected Learning (APCI) program	Math and language (Spanish)	180 min./week during school	Personalized curriculum based on screening test	No, but questions depend on screening test	Not specified	No effect in math or language	Not specified
Lai et al. (2012)	RCT	Grade 3, 57 public rural schools, Qinghai, China	Not specified	Language (Mandarin)	Two 40-min. mandatory sessions/week during lunch breaks or after school; teams of 2 children	National curriculum, reinforces material taught that week	No, same questions for all students	No, if students had a question, they could discuss it with their teammate, but not the teacher	No effect in language and 0.23σ in math	Not specified
Lai et al. (2013)	RCT	Grades 3 and 5, 72 rural boarding schools, Shaanxi, China	Not specified	Math	Two 40-min. mandatory sessions/week after school; teams of 2 children	National curriculum, reinforces material taught that week	No, same questions for all students	No, if students had a question, they could discuss it with their teammate, but not the teacher	0.12σ in language, across both grades	Not specified

Mo et al. (2014b)	RCT	Grades 3 and 5, 72 rural schools, Shaanxi, China	Not specified	Math	Two 40-min. mandatory sessions/week during computer lessons; teams of 2 children	National curriculum, reinforces material taught that week	No, same questions for all students	No, if students had a question, they could discuss it with their teammate, but not the teacher	0.18 σ in math	USD 9439 in total for 1 year
Mo et al. (2014a)	RCT	Grades 3 and 5, 72 rural schools, Shaanxi, China	Not specified	Math	Two 40-min. mandatory sessions/week during computer lessons; teams of 2 children	National curriculum, reinforces material taught that week	No, same questions for all students	No, if students had a question, they could discuss it with their teammate, but not the teacher	Phase 1: no effect in math; Phase 2: 0.3 σ in math	USD 9439 in total for 1 year
Lai et al. (2015b)	RCT	Grade 3, 43 migrant schools, Beijing, China	Not specified	Math	Two 40-min. mandatory sessions/week during lunch breaks or after school	National curriculum, reinforces material taught that week	No, same questions for all students	No, if students had a question, they could discuss it with their teammate, but not the teacher	0.15 σ in math and no effect in language	USD 7.9-8.8 per child for 6 months
Mo et al. (2016)	RCT	Grade 5, 120 schools, Qinghai, China	Not specified	Language (English)	Version 1: Two 40-min. mandatory sessions/week during regular computer lessons; Version 2: English lessons (also optional during lunch or other breaks); Both: teams of 2 children	National curriculum, reinforces material taught that week	Version 1: No feedback during regular computer lessons; Version 2: feedback from teachers during English lessons	Version 1: if students had a question, they could discuss it with their teammate, but not the teacher; Version 2: feedback from English teacher	Version 1: 0.16 σ in language; Version 2: no effect in language	Version 1: RMB 32.09 (USD 5.09) per year; Version 2: RMB 24.42 (USD 3.87) per year

Wise and Olson (1995)	RCT	Grades 2-5, 4 public schools in Boulder, Colorado, United States	Reading with Orthographic and Segmented Speech (ROSS) programs	Language and reading (English)	Both versions: 420 total min., in 30- and 15-min. sessions; teams of 3 children	Not specified	No, but harder problems introduced only once easier problems solved correctly; also in Version 2, teachers explained questions answered incorrectly	No, but students can request help when they do not understand a word	Positive effect on the Lindamond Test of Auditory Conceptualization (LAC), Phoneme Deletion test and Nonword Reading (ESs not reported); no effect on other language and reading domains	Not specified
Morgan and Ritter (2002)	RCT	Grade 9, 4 public schools in Moore Independent School District, Oklahoma, United States	Cognitive Tutor - Algebra I	Math	Not specified	Not specified	Not specified	Not specified	Positive effect (ES not reported) in math	Not specified
Rouse and Krueger (2004)	RCT	Grades 4-6, 4 public schools in urban district in northeast United States	Fast For Word (FFW) programs	Language and reading (English)	90-100 min./day during lessons ("pull-out") or before/after school, 5 days a week, for 6-8 weeks	Not specified	No, but harder problems introduced only once easier problems solved correctly	Not specified	No effect on Reading Edge test, Clinical Evaluation of Language Fundamentals 3rd Edition (CELF-3-RP), Success For All (SFA) test, or State Reading Test	USD 30,000 for a 1-year license for 30 computers, plus USD 100 per site for professional training

Dynarski et al. (2007)	RCT	Grades 4-6, 4 public schools in urban district in northeast United States	Fast For Word (FFW) programs	Language and reading (English)	90-100 min./day during lessons ("pull-out") or before/after school, 5 days a week, for 6-8 weeks	Not specified	No, but harder problems introduced only once easier problems solved correctly	Not specified	No effect on Reading Edge test, Clinical Evaluation of Language Fundamentals 3rd Edition (CELF-3-RP), Success For All (SFA) test, or State Reading Test	USD 30,000 for a 1-year license for 30 computers, plus USD 100 per site for professional training
		Grade 4, 43 public schools in 11 school districts, United States	Leapfrog, Read 180, Academy of Reading, Knowledgebox	Reading (English)	Varies by product, but 70% used them during class time; 25% used them before school, during lunch breaks, or time allotted to other subjects; and 6% of teachers used them during both	Not specified	Not specified, but all four products automatically created individual "learning paths" for each student	Not specified, but all four products provided immediate feedback to students; one provided feedback of mastery; two provided feedback on diagnostics	No effect in reading	USD 18 to USD 184 per student year (depending on the product)
		Grade 6, 28 public schools in 10 school districts, United States	Larson Pre-Algebra, Achieve Now, iLearn Math	Math	Varies by product, but 76% used them during class time; 11% used them before school, during lunch breaks, or time allotted to other subjects; and 13% of teachers used them during both	Not specified	Not specified, but all three products automatically created individual "learning paths" for each student	Not specified, but all three products provided immediate feedback to students; one provided feedback of mastery; two provided feedback on diagnostics	No effect in math	USD 9 to USD 30 per student year (depending on the product)

Algebra I, 23 public schools in 10 school districts, United States	Cognitive Tutor - Algebra I, PLATO Algebra, Larson Algebra	Math	Varies by product, but 94% used them during class time; and 6% of teachers used them during both	Not specified	Not specified, but two products automatically created individual "learning paths" for each student	Not specified, but all three products provided immediate feedback to students; two provided feedback of mastery; two provided feedback on diagnostics	No effect in math	USD 7 to USD 30 per student year year (depending on the product)		
Barrow et al. (2009)	RCT	Grades 8, 10	I Can Learn	Math	Not specified	National Council of Teachers of Mathematics (NCTM) standards and district course objectives	No, but students who do not pass comprehensive tests repeat lessons until they pass them	Not specified	0.17σ in math	30-seat lab costs USD 100,000, with an additional USD 150,000 for pre-algebra, algebra, and classroom management software
Borman et al. (2009)	RCT	Grades 2 and 7, 8 public schools in Baltimore, Maryland, United States	Fast For Word (FFW) Language	Language and reading (English)	100 min./day, five days a week, for four to eight weeks, during lessons ("pull-out")	Not specified	No, all children start at the same basic level and advance only after attaining a pre-determined level of proficiency	Not specified	Grade 2: no effect in language or reading; Grade 7: no effect in language or reading	Not specified
Cam-puzano et al. (2009)	RCT	Grade 1, 12 public schools in 2 school districts, United States	Destination Reading - Course 1	Reading (English)	20 min./day, twice a week, during school	Not specified	Not specified	Not specified	No effect in reading	USD 78 per student per year
		Grade 1, 12 public schools in 3 school districts, United States	Headsprout	Reading (English)	30 min./day, three times a week, during school	Not specified	Not specified	Not specified	0.01 SDs in reading (p<0.05)	USD 146 per student per year

Grade 1, 8 public schools in 3 school districts, United States	PLATO Focus	Reading (English)	15-30 min./day (frequency per week not specified)	Not specified	No, but teachers can choose the order and difficulty level for activities	Not specified	No effect in reading	USD 351 per student per year
Grade 1, 13 public schools in 3 school districts, United States	Waterford Early Reading Program - Levels 1-3	Reading (English)	17-30 min./day, three times a week, during school	Not specified	Not specified	Not specified	No effect in reading	USD 223 per student per year
Grade 4, 15 public schools in 4 school districts, United States	Academy of Reading	Reading (English)	25 min./day, three or more days a week, during school	Not specified	Not specified	Not specified	No effect in reading	USD 217 per student per year
Grade 4, 19 public schools in 4 school districts, United States	LeapTrack	Reading (English)	15 min./day, three to five days a week, during school	Not specified	No, but diagnostic assessments determine "learning path" for each student	Not specified	0.09 σ in reading	USD 154 per student per year
Grade 6, 13 public schools in 3 school districts, United States	PLATO Achieve Now - Mathematics Series 3	Math	30 min./day, four days a week, for at least 10 weeks, during school	Not specified	No, but diagnostic assessment determines which activities students should attempt	Not specified	No effect in math	USD 36 per student per year
Grade 6, 13 public schools in 5 school districts, United States	Larson Pre-Algebra	Math	Varies according to the number of topics/weeks in the course, but recommended at least one a week	Not specified	Not specified	Not specified	No effect in math	USD 15 per student per year

Algebra I, 11 public schools in 4 school districts, United States	Cognitive Tutor - Algebra I	Math	Two days a week (plus textbook three days a week)	Not specified	Not specified	Not specified	No effect in math	USD 69 per student per year
Algebra I, 12 public schools in 5 school districts, United States	Larson Algebra I	Math	Varies according to the number of topics/weeks in the course, but recommended at least one a week	Not specified	Not specified	Not specified	No effect in math	USD 13 per student per year
Rockoff (2015)	RCT	Math	Not specified	No, activities sourced from publishers, software providers, and other educational groups	Yes, "learning algorithm" draws on students' performance on each lesson and recommends a "playlist" for each student; at the end of the day, students take a "playlist update"	No, but possibility to get feedback from live reinforcement of prior lessons, live tutoring, small group collaboration, virtual live instruction, and virtual live tutoring	No effect on New York State Math Test or Northwest Evaluation Association (NWEA) test	Not specified

Appendix C Mindspark software

This appendix provides a more detailed description of the working of the Mindspark computer-assisted learning (CAL) software, and specifics of how it was implemented in the after-school Mindspark centers evaluated in our study.

C.1 Computer training

The first time that students log into the Mindspark software, they are presented with an optional routine (taking 10-15 minutes) designed to familiarize them with the user interface and exercises on math or language.

C.2 Diagnostic test

After the familiarization routine, students are presented with diagnostic tests in math and Hindi which are used by the Mindspark platform to algorithmically determine their initial achievement level (at which instruction will be targeted). Tests contain four to five questions per grade level in each subject. All students are shown questions from grade 1 up to their grade level. However, if students answer at least 75% of the questions for their corresponding grade level correctly, they can be shown questions up to two grade levels above their own.³³ If they answer 25% or less of the questions for one grade level above their actual grade, the diagnostic test shows no more questions. Initial achievement levels determined by the Mindspark system on the basis of these tests are only used to customize the first set of content that students are provided. Further customization is based on student performance on these content modules and does not depend on their performance on the initial diagnostic test (which is only used for initial calibration of each student’s learning level).

C.3 Math and Hindi content

Mindspark contains a number of activities that are assigned to specific grade levels, based on analyses of state-level curricula. All of the items are developed by EI’s education specialists. The Mindspark centers focus on a specific subject per day: there are two days assigned to math, two days assigned to Hindi, one day assigned to English, and a “free” day, in which students can choose a subject.

Math and Hindi items are organized differently. In math, “topics” (e.g., whole number operations) are divided into “teacher topics” (e.g., addition), which are divided into “clusters” (e.g., addition in a number line), which are divided into “student difficulty levels” (SDLs) (e.g., moving from one place to another on the number line), which are in turn divided into questions (e.g., the same exercise with slightly different numbers). The Mindspark software

³³For example, a grade 4 student will always see questions from grade 1 up to grade 4. However, if he/she answers over 75% of grade 4 questions correctly, he/she will be shown grade 5 questions; and if he/she answers over 75% of grade 5 questions correctly, he/she will be shown grade 6 questions.

currently has 21 topics, 105 teacher topics and 550 clusters. The organization of math content reflects the mostly linear nature of math learning (e.g., you cannot learn multiplication without understanding addition). This is also why students must pass an SDL to move on to the next one, and SDLs always increase in difficulty.

In Hindi, there are two types of questions: “passages” (i.e., reading comprehension questions) and “non-passages” (i.e., questions not linked to any reading). Passage questions are grouped by grades (1 through 8), which are in turn divided into levels (low, medium, or high). Non-passage questions are grouped into “skills” (e.g., grammar), which are divided into “sub-skills” (e.g., nouns), which are in turn divided into questions (e.g., the same exercise with slightly different words). The Mindspark software currently has around 330 passages (i.e., 20 to 50 per grade) linked to nearly 6,000 questions, and for non-passage questions, 13 skills and 50 sub-skills, linked to roughly 8,200 questions. The Hindi content is organized in this way because language learning is not as linear as math (e.g., a student may still read and comprehend part of a text even if he/she does not understand grammar or all the vocabulary words in it). As a result there are no SDLs in Hindi, and content is not necessarily as linear or clearly mapped into grade-level difficulty as in math.

The pedagogical effectiveness of the language-learning content is increased by using videos with same-language subtitling (SLS). The SLS approach relies on a “karaoke” style and promotes language learning by having text on the screen accompany an audio with on-screen highlighting of the syllable on the screen at the same time that it is heard, and has been shown to be highly effective at promoting adult literacy in India (Kothari et al., 2002, 2004). In Mindspark, the SLS approach is implemented by showing students animated stories with Hindi audio alongside subtitling in Hindi to help the student read along and improve phonetic recognition, as well as pronunciation.

C.4 Personalization

C.4.1 Dynamic adaptation to levels of student achievement

In math, the questions within a teacher topic progressively increase in difficulty, based on EI’s data analytics and classification by their education specialists. When a child does not pass a learning unit, the learning gap is identified and appropriate remedial action is taken. It could be leading the child through a step-by-step explanation of a concept, a review of the fundamentals of that concept, or simply more questions about the concept.

Figure C.1 provides an illustration of how adaptability works. For example, a child could be assigned to the “decimal comparison test”, an exercise in which he/she needs to compare two decimal numbers and indicate which one is greater. If he/she gets most questions in that test correctly, he/she is assigned to the “hidden numbers game”, a slightly harder exercise in which he/she also needs to compare two decimal numbers, but needs to do so with as

little information as possible (i.e., so that children understand that the digit to the left of the decimal is the most important and those to the right of the decimal are in decreasing order of importance). However, if he/she gets most of the questions in the decimal comparison test incorrectly, he/she is assigned to a number of remedial activities seeking to reinforce fundamental concepts about decimals.

In Hindi, in the first part, students start with passages of low difficulty and progress towards higher-difficulty passages. If a child performs poorly on a passage, he/she is assigned to a lower-difficulty passage. In the second part, students start with questions of low difficulty in each skill and progress towards higher-difficulty questions. Thus, a student might be seeing low-difficulty questions on a given skill and medium-difficulty questions on another.

C.4.2 Error analysis

Beyond adapting the level of difficulty of the content to that of the student, Mindspark also aims to identify specific sources of conceptual misunderstanding for students who may otherwise be at a similar overall level of learning. Thus, while two students may have the same score on a certain topic (say scoring 60% on fractions), the reasons for their missing the remaining questions may be very different, and this may not be easy for a teacher to identify. A distinctive feature of the Mindspark system is the use of detailed data on student responses to each question to analyze and identify *patterns* of errors in student responses to allow for identifying the precise misunderstanding/misconception that a student may have on a given topic, and to target further content accordingly.

The idea that educators can learn as much (or perhaps more) from analyzing patterns of student errors than from their correct answers has a long tradition in education research (for instance, see (Buswell and Judd, 1925) and (Radatz, 1979) for discussions of the use of “error analysis” in mathematics education). Yet, implementing this idea in practice is highly non-trivial in a typical classroom setting for individual teachers. The power of ‘big data’ in improving the design and delivery of educational content is especially promising in the area of error analysis, as seen in the example below.

Figure C.2 shows three examples of student errors in questions on “decimal comparison”. These patterns of errors were identified by the Mindspark software, and subsequently EI staff interviewed a sample of students who made these errors to understand their underlying misconceptions. In the first example, students get the comparison wrong because they exhibited what EI classifies as “whole number thinking”. Specifically, students believed 3.27 was greater than 3.3 because, given that the integer in both cases was the same (i.e., 3), they compared the numbers to the left of the decimal point (i.e., 27 and 3) and concluded (incorrectly) that since 27 is greater than 3, 3.27 was greater than 3.3.

In the second example, the error cannot be because of the reason above (since 27 is greater than 18). In this case, EI diagnosed the nature of the misconception as “reverse order thinking”. In this case, students know that the ‘hundred’ place value is greater than the ‘ten’ place value, but also believe as a result that the ‘hundred th ’ place value is greater than the ‘tent th ’ place value. Therefore, they compared 81 to 27 and concluded (incorrectly) that 3.18 was greater than 3.27.

Finally, the error in the last example cannot be because of either of the two patterns above (since 27 is less than 39, and 7 is less than 9). In this case, EI diagnosed the nature of the misconception as “reciprocal thinking”. Specifically, students in this case understood that the component of the number to the right of the decimal is a fraction, but they then proceeded to take the reciprocal of the number to the right of the decimal, the way standard fractions are written. Thus, they were comparing $\frac{1}{27}$ to $\frac{1}{39}$ as opposed to 0.27 to 0.39 and as a result (incorrectly) classified the former as greater.

It is important to note that the fraction of students making each type of error is quite small (5%, 4%, and 3% respectively), which would make it much more difficult for a teacher to detect these patterns in a typical classroom (since the sample of students in a classroom would be small). The comparative advantage of the computer-based system is clearly apparent in a case like this, since it is able to analyze patterns from thousands of students, with each student attempting a large set of such comparisons. This enables both pattern recognition at the aggregate level and diagnosis at the individual student-level as to whether a given student is exhibiting that pattern. Consistent with this approach, Mindspark then targets follow-up content based on the system’s classification of the patterns of student errors as seen in Figure C.1 (which also shows how each student would do 30 comparisons in the initial set of exercises to enable a precise diagnosis of misconceptions).

C.5 Feedback

The pedagogical approach favoured within the Mindspark system prioritizes active student engagement at all times. Learning is meant to build upon feedback to students on incorrect questions. Also, most questions are preceded by an example and interactive content that provide step-by-step instructions on how students should approach solving the question.

In math, feedback consists of feedback to wrong answers, through animations or text with voice-over. In Hindi, students receive explanations of difficult words and are shown how to use them in a sentence. The degree of personalization of feedback differs by question: (a) in some questions, there is no feedback to incorrect answers; (b) in others, all students get the same feedback to an incorrect answer; and (c) yet in others, students get different types of feedback depending on the wrong answer they selected.

Algorithms for the appropriate feedback and further instruction that follow a particular pattern of errors are informed by data analyses of student errors, student interviews conducted by EI's education specialists to understand misconceptions, and published research on pedagogy. All decisions of the software in terms of what content to provide after classification of errors are 'hard coded' at this point. Mindspark does not currently employ any machine-learning algorithms (although the database offers significant potential for the development of such tools).

In addition to its adaptive nature, the Mindspark software allows the center staff to provide students with an 'injection' of items on a given topic if they believe a student needs to review that topic. However, once the student completes this injection, the software reverts to the item being completed when the injection was given and relies on its adaptive nature.

Figure C.1: Mindspark adaptability in math

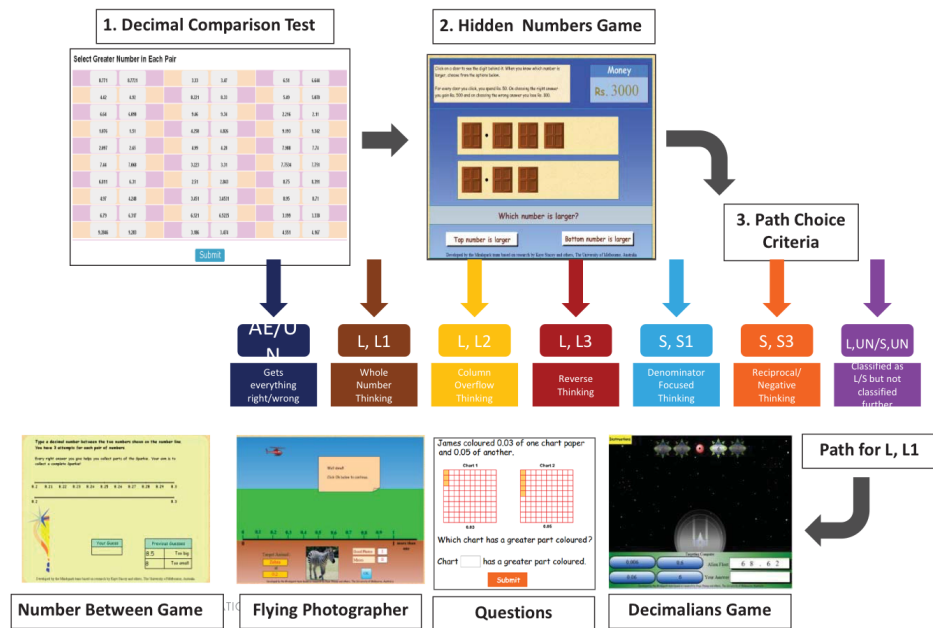
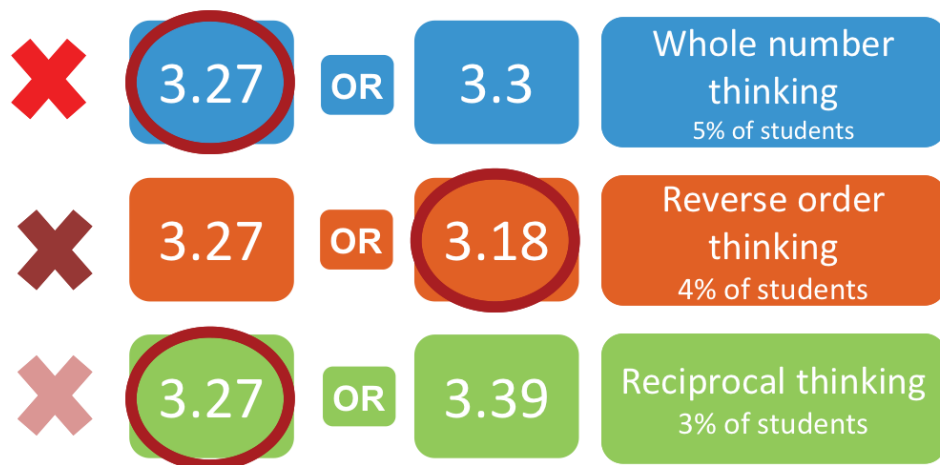


Figure C.2: Student errors in math



Appendix D Test design

D.1 Overview

We measured student achievement, which is the main outcome for our evaluation, using independent assessments in math and Hindi. These tests were administered under the supervision of the research team at both baseline and endline. Here we present details about the test content and development, administration, and scoring.

D.2 Objectives of test design

Our test design was informed by three main objectives. First, was to develop a test which would be informative over a wide range of achievement. Recognizing that students may be much below grade-appropriate levels of achievement, test booklets included items ranging from very basic primary school appropriate competences to harder items which are closer to grade-appropriate standards.

Our secondary objective was to ensure that we were measuring a broad construct of achievement which included both curricular skills and the ability to apply them in simple problems.

Our third, and related, objective was to ensure that the test would be a fair benchmark to judge the actual skill acquisition of students. Reflecting this need, tests were administered using pen-and-paper rather than on computers so that they do not conflate increments in actual achievement with greater familiarity with computers in the treatment group. Further, the items were taken from a wide range of independent assessments detailed below, and selected by the research team without consultation with Education Initiatives, to ensure that the selection of items was not prone to “teaching to the test” in the intervention.

D.3 Test content

We aimed to test a wide range of abilities. The math tests range from simple arithmetic computation to more complex interpretation of data from charts and framed examples as in the PISA assessments. The Hindi assessments included some “easy” items such as matching pictures to words or Cloze items requiring students to complete a sentence by supplying the missing word. Most of the focus of the assessment was on reading comprehension, which was assessed by reading passages of varying difficulty and answering questions that may ask students to either retrieve explicitly stated information or to draw more complex inferences based on what they had read. In keeping with our focus on measuring functional abilities, many of the passages were framed as real-life tasks (e.g. a newspaper article, a health immunization poster, or a school notice) to measure the ability of students to complete standard tasks.

In both subjects, we assembled the tests using publicly available items from a wide range of research assessments. In math, the tests drew upon items from the Trends in Mathematics and Science Study (TIMSS) 4th and 8th grade assessments, OECD’s Programme for International Student Assessment (PISA), the Young Lives student assessments administered in four countries including India, the Andhra Pradesh Randomized Studies in Education (APRESt), the India-based Student Learning Survey (SLS) and Quality Education Study (QES); these collectively represent some of the most validated tests in the international and the Indian context.

In Hindi, the tests used items administered by Progress in International Reading Literacy Study (PIRLS) and from Young Lives, SLS and PISA. These items, available in the public domain only in English were translated and adapted into Hindi.

D.4 Test booklets

We developed multiple booklets in both baseline and endline for both subjects. In the baseline assessment, separate booklets were developed for students in grades 4-5, grades 6-7 and grades 8-9. In the endline assessment, given the very low number of grades 4-5 students in our study sample, a single booklet was administered to students in grades 4-7 and a separate booklet for students in grades 8-9. Importantly, there was substantial overlap that was maintained between the booklets for different grades and between the baseline and endline assessments. This overlap was maintained across items of all difficulty levels to allow for robust linking using IRT. Table D.1 presents a break-up of questions by grade level of difficulty in each of the booklets at baseline and endline.

Test booklets were piloted prior to baseline and items were selected based on their ability to discriminate achievement among students in this context. Further, a detailed Item analysis of all items administered in the baseline was carried out prior to the finalization of the endline test to ensure that the subset of items selected for repetition in the endline performed well in terms of discrimination and were distributed across the ability range in our sample. Table D.2 presents the number of common items which were retained across test booklets administered.

D.5 Test scoring

All items administered were multiple-choice questions, responses to which were marked as correct or incorrect dichotomously. The tests were scored using Item Response Theory (IRT) models.

IRT models specify a relationship between a single underlying latent achievement variable (“ability”) and the probability of answering a particular test question (“item”) correctly. While standard in the international assessments literature for generating comparative test scores, the use of IRT models is much less prevalent in the economics of education literature

in developing countries (for notable exceptions, see Das and Zajonc 2010, Andrabi et al 2011, Singh 2015). For a detailed introduction to IRT models, please see Van der Linden and Hambleton (1997) and Das and Zajonc (2010).

The use of IRT models offers important advantages in an application such as ours, especially in comparison to the usual practice of presenting percentage correct scores or normalized raw scores. First, it allows for items to contribute differentially to the underlying ability measure; this is particularly important in tests such as ours where the hardest items are significantly more complex than the easiest items on the test.

Second, it allows us to robustly link all test scores on a common metric, even with only a partially-overlapping set of test questions, using a set of common items between any two assessments as “anchor” items. This is particularly advantageous when setting tests in samples with possibly large differences in mean achievement (but which have substantial common support in achievement) since it allows for customizing tests to the difficulty level of the particular sample but to still express each individual’s test score on a single continuous metric. This is particularly important in our application in enabling us to compute business-as-usual value-added in the control group.³⁴

Third, IRT models also offer a framework to assess the performance of each test item individually which is advantageous for designing tests that include an appropriate mix of items of varying difficulty but high discrimination.

We used the 3-parameter logistic model to score tests. This model posits the relationship between underlying achievement and the probability of correctly answering a given question as a function of three item characteristics: the difficulty of the item, the discrimination of the item, and the pseudo-guessing parameter. This relationship is given by:

$$P_g(\theta_i) = c_g + \frac{1 - c_g}{1 + \exp(-1.7 \cdot a_g \cdot (\theta_i - b_g))} \quad (4)$$

where i indexes students and g indexes test questions. θ_i is the student’s latent achievement (ability), P is the probability of answering question g correctly, b_g is the difficulty parameter and a_g is the discrimination parameter (slope of the ICC at b). c_g is the pseudo-guessing parameter which takes into account that, with multiple choice questions, even the lowest ability can answer some questions correctly.

Given this parametric relationship between (latent) ability and items characteristics, this relationship can be formulated as a joint maximum likelihood problem which uses the matrix of $N \times M$ student responses to estimate $N + 3M$ unknown parameters. Test scores were generated

³⁴IRT scores are only identified up to a linear transformation. Without explicitly linking baseline and endline scores, the constant term in our value-added regressions (which we interpret as value-added in the control group) would have conflates the arbitrary linear transformation and value-added in the control group.

using the OpenIRT software for Stata written by Tristan Zajonc. We use maximum likelihood estimates of student achievement in the analysis which are unbiased individual measures of ability (results are similar when using Bayesian expected a posteriori scores instead).

D.6 Empirical distribution of test scores

Figure D.1 presents the percentage correct responses in both math and Hindi for baseline and endline. It shows that the tests offer a well-distributed measure of achievement with few students unable to answer any question or to answer all questions correctly. This confirms that our achievement measures are informative over the full range of student achievement in this setting.

Figure D.2 presents similar graphs for the distribution of IRT test scores. Note that raw percent correct scores in Figure D.1 are not comparable over rounds or across booklets because of the different composition of test questions but the IRT scores used in the analysis are.

D.7 Item fit

The parametric relationship between the underlying ability and item characteristics is assumed, in IRT models, to be invariant across individuals (in the psychometrics literature, referred to as no differential item functioning). An intuitive check for the performance of the IRT model is to assess the empirical fit of the data to the estimated item characteristics.

Figure D.2 plots the estimated Item Characteristic Curve (ICC) for each individual item in math and Hindi endline assessments along with the empirical fit for treatment and control groups separately. The fit of the items is generally quite good and there are no indications of differential item functioning (DIF) between the treatment and control groups. This indicates that estimated treatment effects do not reflect a (spurious) relationship induced by a differential performance of the measurement model in treatment and control groups.

Figure D.1: Distribution of raw percentage correct scores

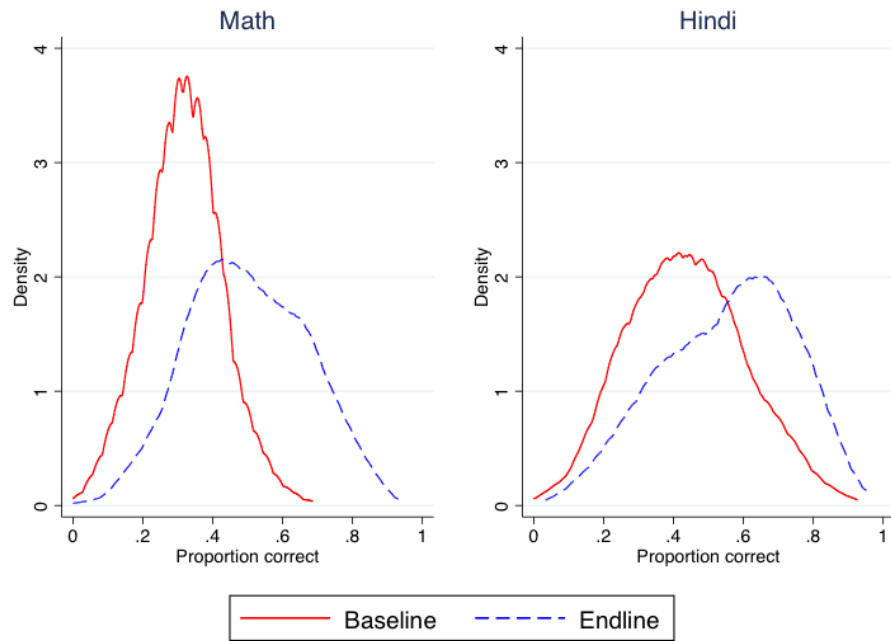


Figure D.2: Distribution of IRT scores, by round and treatment status

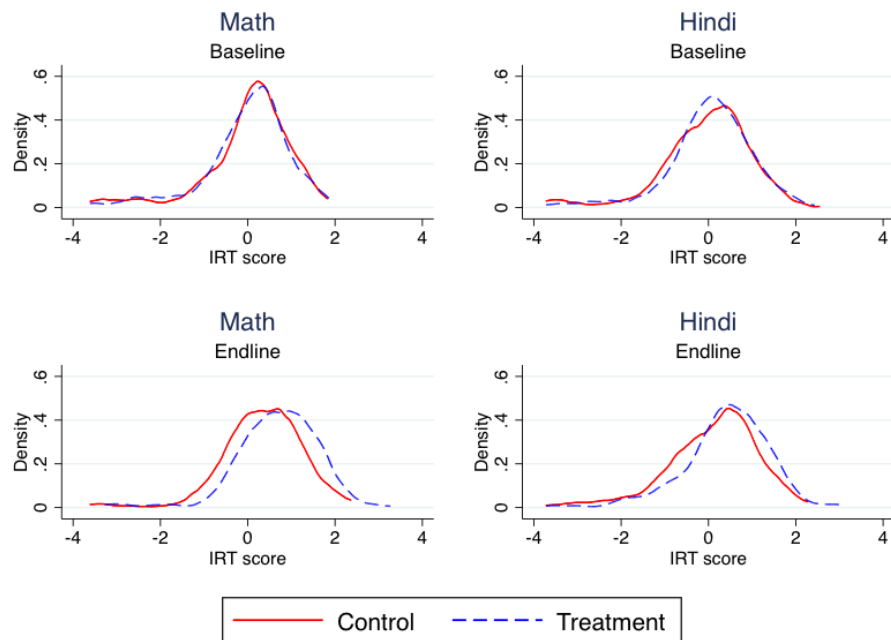
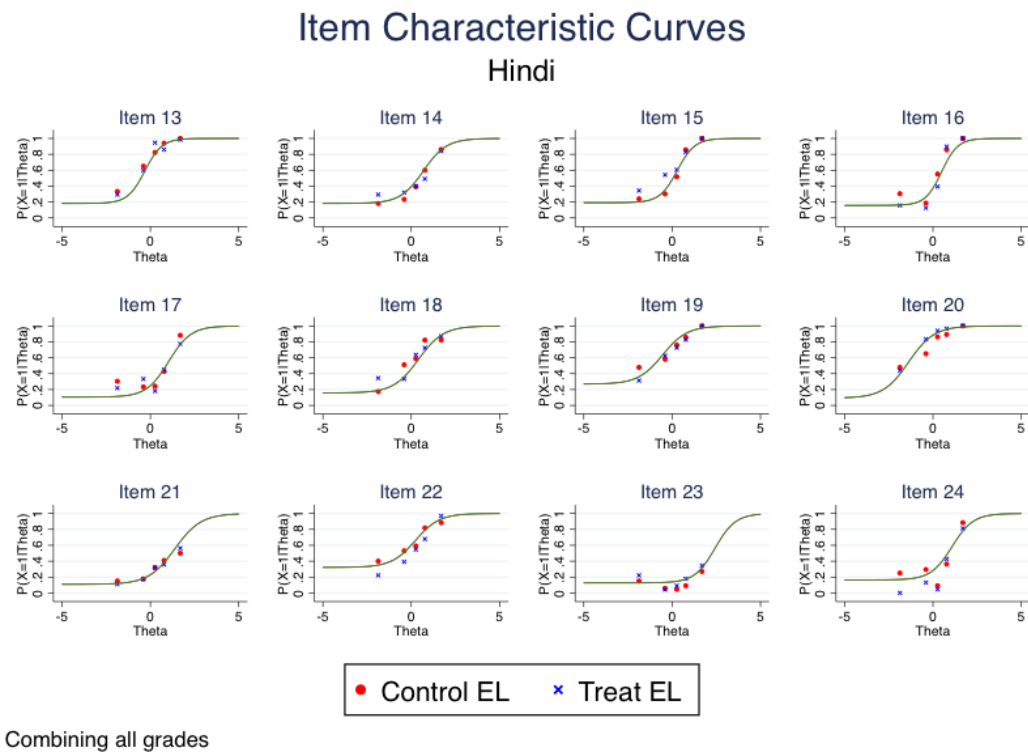
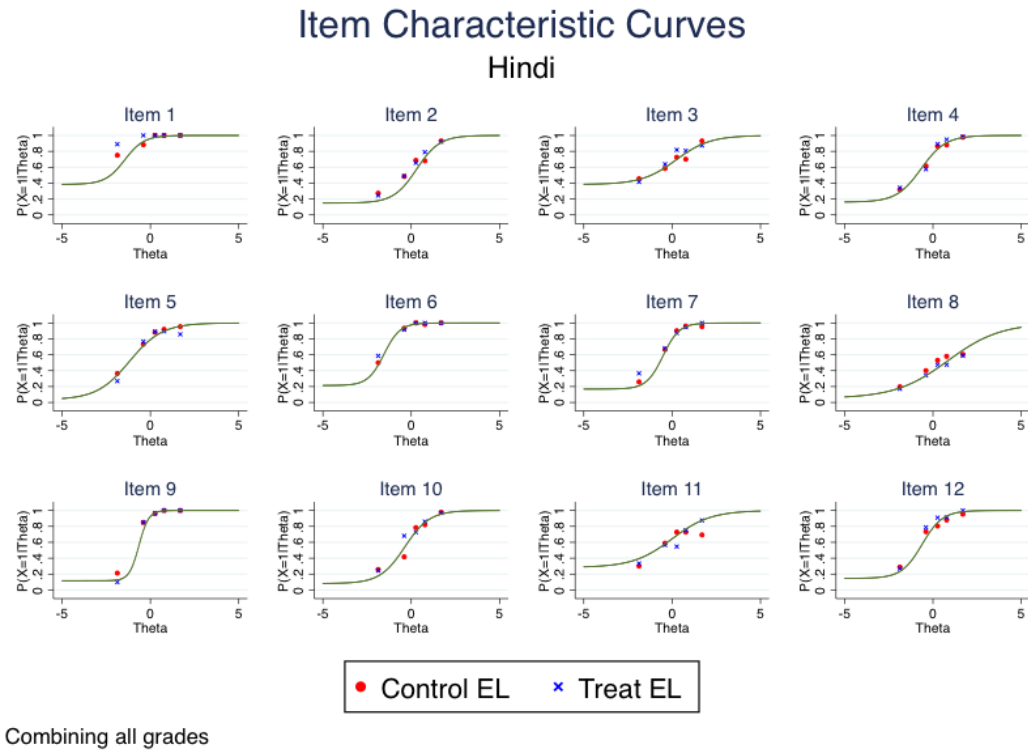
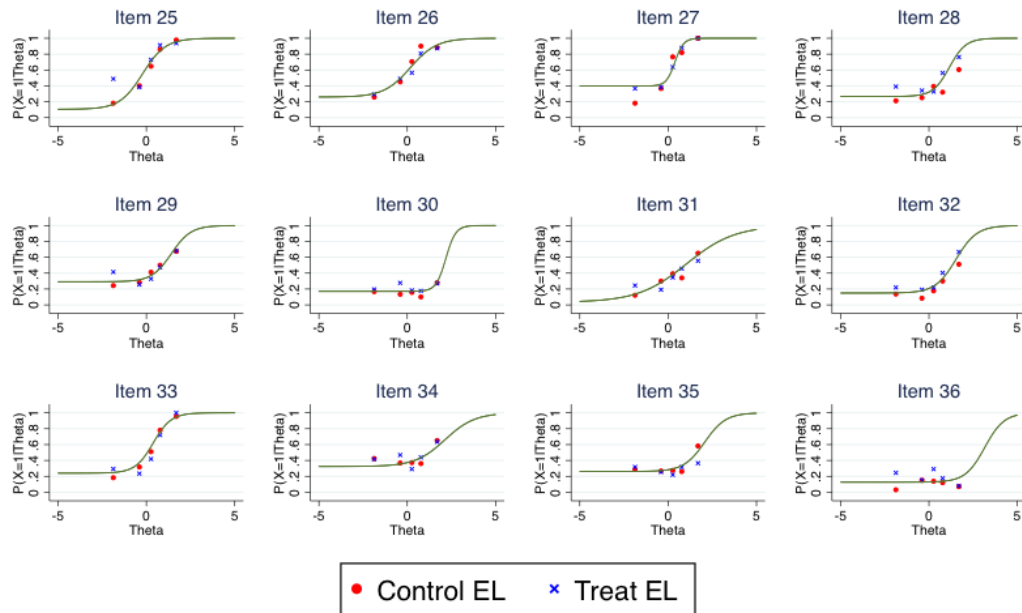


Figure D.3: Item Characteristic Curves: Hindi



Item Characteristic Curves

Hindi

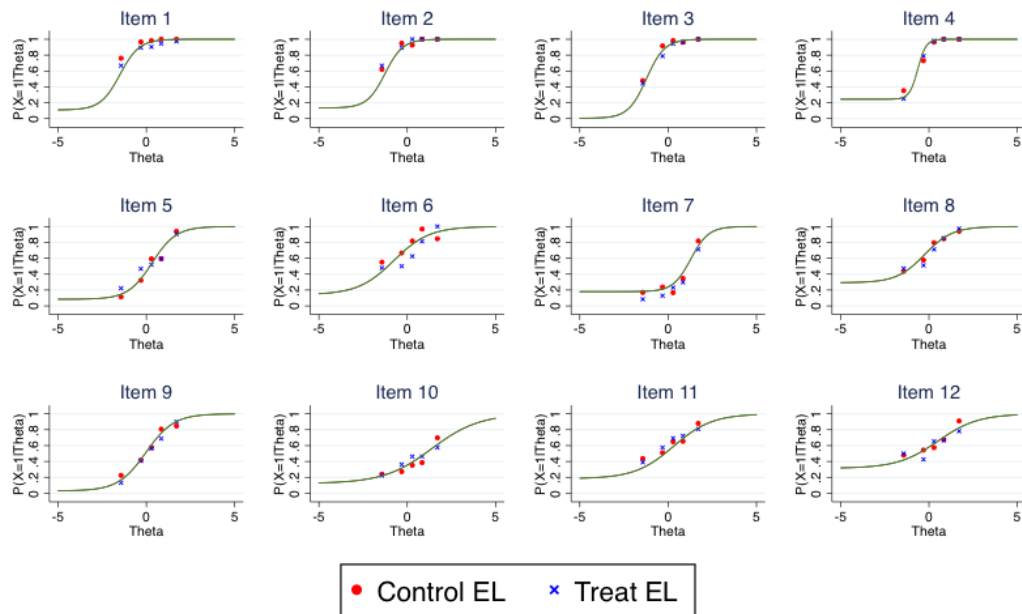


Combining all grades

Figure D.4: Item Characteristic Curves: Math

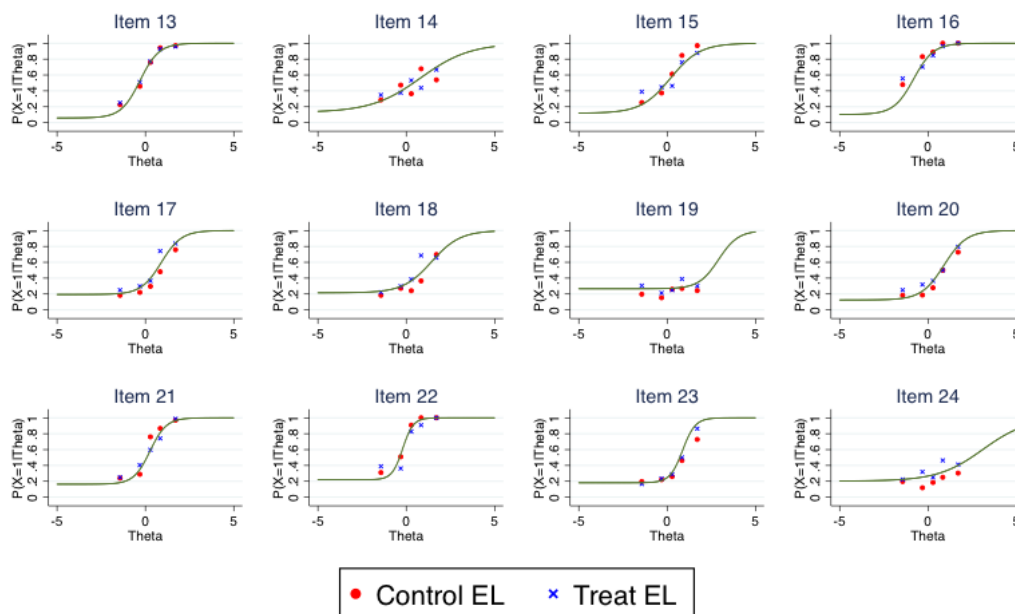
Item Characteristic Curves

Mathematics



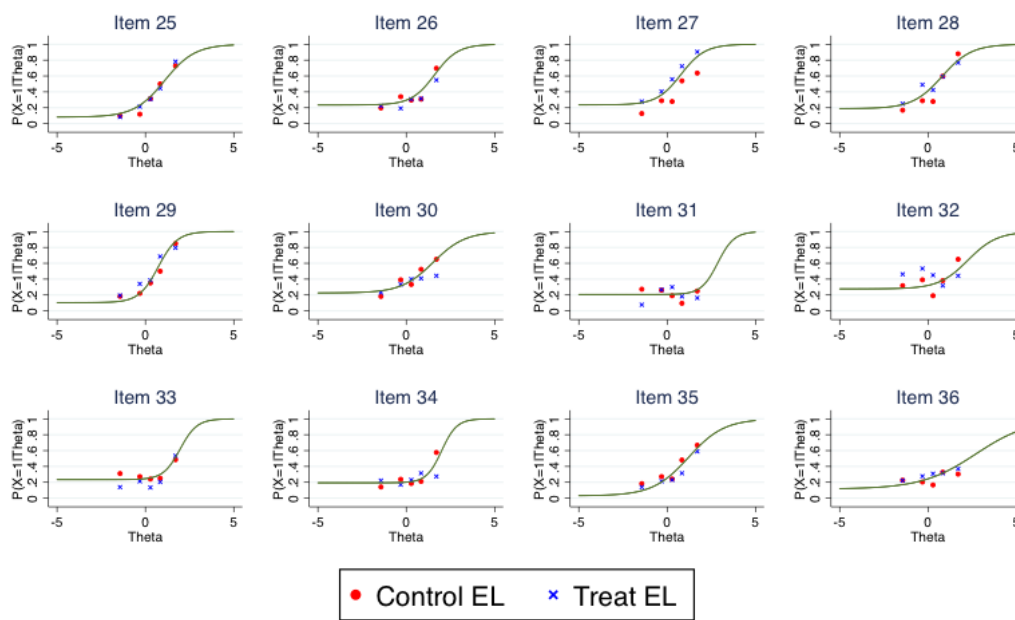
Combining all grades

Item Characteristic Curves Mathematics



Combining all grades

Item Characteristic Curves Mathematics



Combining all grades

Table D.1: Distribution of questions by grade-level difficulty across test booklets

		Booklets				
		Baseline			Endline	
		Math				
		G4-5	G6-7	G8-9	G4-7	G8-9
Number of questions	G2	2	0	0	2	0
at each grade level	G3	14	6	4	6	6
	G4	13	7	4	9	8
	G5	4	10	3	10	10
	G6	1	10	10	5	6
	G7	1	2	11	2	3
	G8	0	0	3	0	2
		Hindi				
		G4-5	G6-7	G8-9	G4-7	G8-9
Number of questions	G2	5	2	1	1	0
at each grade level	G3	3	4	2	1	1
	G4	7	3	3	8	8
	G5	8	7	2	5	6
	G6	0	2	3	11	11
	G7	0	5	9	0	4
	G8	7	7	7	4	0
	G9	0	0	3	0	0

Note: Each cell presents the number of questions by grade-level of content across test booklets. The tests were designed to capture a wide range of student achievement and thus were not restricted to grade-appropriate items only. The grade-level of test questions was established ex-post with the help of a curriculum expert.

Table D.2: Distribution of common questions across test booklets

Math				
	BL G6-7	BL G8-9	EL G4-7	EL G8-9
BL G4-5	16	10	14	14
BL G6-7		15	10	10
BL G8-9			7	7
EL G4-7				31

Hindi				
	BL G6-7	BL G8-9	EL G4-7	EL G8-9
BL G4-5	18	10	11	9
BL G6-7		17	13	13
BL G8-9			9	8
EL G4-7				24

Note: Each cell presents the number of questions in common across test booklets. Common items across booklets are used to anchor IRT estimates of student achievement on to a common metric.